

Fungal Endophyte Community Analysis — LOT2 (Peru)

Analysis environment: R 4.6.1 | Generated on 2026-10-06

Overview

This project analyses the fungal endophyte communities isolated from three substrates collected in **Peru** (LOT2 sampling campaign):

Substrate	Isolates (pooled)	Sample-level replicates (units)
Ficus leaves	264	10
Ficus wood	167	12
Lauraceae leaves	219	8

Sampling design: what was sampled, and the assumptions behind it

This section records assumptions supplied by the data owner, not facts derivable from the files. They are written out so collaborators can check the reasoning, and because several of them change how the results must be read. If any is wrong, the analyses affected are named underneath it.

1. Two individual trees were sampled — one of each host. A strangler *Ficus* growing on a single Lauraceae host tree. There is no tree-level replication, and no tree identifier in the raw files; `parse_LOT2.R` assigns `tree_id` from the substrate.

Consequence. The **inferential unit is the individual tree, not the host species**. *Ficus* leaves vs *Ficus* wood is a genuine within-tree tissue contrast. But *Ficus* vs Lauraceae compares **one tree with one other tree**, so every “host effect” below describes these two individuals. Sampling units are treated as replicates by PERMANOVA, which is pseudoreplication at the species level: the p-values are real for the trees sampled and cannot be generalised to *Ficus* vs Lauraceae as taxa. All host-level statements are phrased accordingly.

2. The two trees are not independent. The *Ficus* is a strangler growing **on** the Lauraceae, so the two root systems, canopies and microclimates are physically interlocked and share a continuous surface for fungal dispersal.

Consequence. Any host difference is a difference between two *interlocked* individuals, which is a conservative setting for detecting host effects (shared exposure should erode differences, not create them) but rules out treating them as independent samples of two species.

3. The Lauraceae trunk was never sampled — it was inaccessible, completely encased by the strangling *Ficus*. All Lauraceae material came from eight exposed branches (S1–S8), all in **zone 6**.

Consequence. `parse_LOT2.R` assigns `position = "Branch"` to all Lauraceae, so every trunk-vs-branch contrast in this report is a *Ficus* contrast. An earlier `lauraceae_leaves_trunk_vs_branch_*` analysis, built when some Lauraceae branches were wrongly recorded in zone 5 and read as trunk, compared branch against branch and has been removed (see **Corrections to the sampling units** below).

4. Whether the Lauraceae material is leaves or branch wood is still unconfirmed. `LOT2_pooled_counts.xlsx` labels the column “Lauraceae leaves”; the samples workbook has a sheet titled “66. Fungi-Endo wood (Host)”. The data owner indicates the material came from branches, without settling leaf vs wood. The analyses retain the label **Lauraceae leaves**.

Consequence. **The one conclusion that depends on this is the tissue-type claim.** If the Lauraceae material is leaves, the finding “the two leaf substrates resemble each other more than either resembles wood” stands and tissue type is the dominant split. If it is branch wood, that grouping is wrong and the pattern would have to be re-read as *Ficus* vs Lauraceae. Everything else — diversity, orientation, zone — is unaffected. **This should be resolved before the tissue-type interpretation is published.**

5. Each row is an independent colony; no de-replication is required. Repeated isolates of the same genotype within a sampling unit could be either independent colonisations or one colony subsampled. The rule given is that a **shared Hofstetter-culture code marks subsamples of a single colony**. The parser checks this: all **650 isolates carry 650 distinct culture codes**, so no collapsing is needed and abundances are counts of independent colonies.

Consequence. None — but the check now runs on every pipeline execution and will warn if a future data version reuses a code.

6. Only the first sheet of each workbook is authoritative. The remaining sheets (per-substrate extracts of 264 / 167 / 20 isolates, and the sequence sheet) are working material. The pipeline reads sheet 1 for both files, plus the sequence sheet Feuil2 for OTU clustering.

Corrections to the sampling units

The data curator checked the sampling design with the field team (LoT) and found two labelling errors in L0T2_samples.xlsx. Both are corrected by parse_L0T2.R when the workbook is read. The delivered file is left unchanged, and each correction does nothing once a corrected workbook is issued.

- 1. All Lauraceae branches are in zone 6.** The workbook recorded 53 Lauraceae isolates (all of S7, most of S8 and one isolate of S4) as zone 5, copied from a field file that later proved wrong. A sampling unit is *substrate* × *zone* × *unit*, so S4 and S8 were each split into a zone-5 and a zone-6 unit. Earlier versions of this report therefore counted **10 Lauraceae units instead of 8**, including a one-isolate unit (Lauraceae leaves__Z5__S4), and left some Lauraceae units out of the zone-6 comparison (E4).
- 2. Ficus trunk wood is numbered S11-S15**, not S9-S13. The rename keeps the trunk apart from canopy branches S9-S10, where only leaves were sampled. 3 zone-4 trunk isolates still carried the old label S10, which split zone 4 into two units. They now belong to S12, so the *Ficus* trunk has **exactly one sampling unit per zone** (S11 = zone 5 ... S15 = zone 1).

The corrected data have **30 sampling units** (previously 33), and every one of them now matches the sampling design. The design has 13 *Ficus* wood units, but no isolates are recorded for branch S4 wood, so 12 of them contribute data.

Sampling design

Samples were collected at different **tree zones** (heights), zone 1 lowest to zone 6 highest. On the *Ficus*, zones 1-5 are trunk and zone 6 is canopy branch. The Lauraceae trunk was inaccessible (assumption 3 above), so all Lauraceae material comes from **exposed branches in zone 6**.

Substrate	Tree	Zones	Position	Sampling units
Ficus leaves	Ficus (strangler)	6	Canopy branch (S1-S10)	10
Ficus wood	Ficus (strangler)	1-6	Trunk, zones 1-5 (S11-S15) + branch, zone 6 (S1-S8)	12 (of 13 sampled)
Lauraceae leaves	Lauraceae (host)	6	Exposed branch only (S1-S8)	8

Because the Lauraceae contributes no trunk material, every trunk-vs-branch contrast in this report is a **Ficus** contrast. And because each trunk zone is a single sampling unit, the zone analyses have **no replication within any trunk zone** (see Section E).

Sampling orientation Each sampled branch (and, where recorded, trunk face) also carries a **compass orientation**. This information only became usable once the field team reconciled two independent labelling systems: the colour codes written down by the climbers in the field, and the branch

numbering entered into the project database. The two disagreed (notably a blue/turquoise colour clash, where blue should have been reserved for the Lauraceae, and disputed collection zones for the Lauraceae branches, since settled as zone 6), so earlier versions of `L0T2_samples.xlsx` had an empty orientation column.

The re-issued workbook settles that correspondence and adds it as a reconciled column, which the analyses in **Section F** use. Two clean-up rules are applied when reading it:

- Some bearings still carry a replicate index (N1-N4, NW1-NW4). The digit identifies **which branch**, not which direction, so it is stripped: N3 becomes N.
- Material marked ?, and all Ficus trunk wood (for which no orientation was ever recorded), is treated as **unresolved** and excluded from the orientation analyses.

This leaves **519 of 650 isolates** (80%) with a usable bearing; **131** are unresolved. A single sampling unit can span more than one bearing, so the orientation analyses group isolates by **substrate x zone x unit x bearing**, giving **25 orientation-level sampling units** rather than the 30 used elsewhere.

Handling of uncertain taxonomy (*Incertae sedis*)

Many isolates cannot be confidently named at every taxonomic rank. Entries flagged NA, "?", "NO" or "*incertae sedis*" are treated as ***Incertae sedis*** ("of uncertain placement") and are handled **rank by rank**:

- They are **shown** — as a single pooled *Incertae sedis* category — **in the abundance bar charts, the relative-abundance bars and the pie charts**, so that the full isolate count is never hidden. It is always drawn the same way: a neutral grey that no taxon uses, placed last (top of each stacked bar, last slice of each pie, top row of each bar chart), with its share of isolates printed on the chart.
- They are **excluded from every diversity index, rank-abundance curve, Venn diagram and multivariate test**. A single large, shared "unknown" bin behaves like a taxon that is common everywhere: it inflates apparent overlap between substrates and **flattens the real ecological differences** we are trying to detect.
- Because the exclusion is applied **independently at each rank**, an isolate that is unresolved at *species* level but has a defined *genus*, *family* or *order* still contributes to the analyses run at those higher ranks (see **Section D — Multi-level analyses**).

The sample-level multivariate analyses are run at **ITS-genotype** resolution, where each of the sequenced genotypes is a distinct entity. There is no dominant "unknown" bin at that level, so no isolates are dropped there; the *Incertae sedis* filtering matters only when isolates are grouped into higher taxa.

Input Data

- **L0T2_pooled_counts.xlsx** (first sheet) — Pooled genotype counts per substrate with full taxonomy (284 genotypes)
- **L0T2_samples.xlsx** (first sheet) — Individual isolate records with sampling zone, unit and reconciled branch/trunk orientation (650 isolates)

The workbook also keeps the reconciliation working columns (field colour code, database colour code, pre-reconciliation orientation, field notes). They are read for traceability but only the reconciled orientation column is used. Sheet `Feuil2` holds one ITS sequence per isolate and is used for OTU clustering. Remaining sheets are working material and are not read.

- **L0T2_otu_map.csv** — isolate → OTU assignment produced by `cluster_otus.sh` (142 OTUs at 97% ITS identity). Committed so the pipeline runs without `vsearch`.

The analytical unit

Community analyses use **97% ITS OTUs**, not the BLAST-derived name strings. Those names are not a clustered unit: they over-split (near-identical sequences filed under two spellings, e.g. *guandongensis* / *guangdongensis*) and over-lump (single labels covering “26 spp.”). Clustering the 650 ITS sequences themselves collapses 284 name labels into **142 OTUs**, and materially improves the data:

	Name labels	97% OTUs
Taxa	284	142
Good’s coverage	70%	74.9-88.6%
Sample pairs sharing no taxon	51%	37%

`cluster_otus.sh` regenerates the mapping (needs `vsearch`); it is deterministic and only needs re-running if the sequences change.

Input integrity checks

`parse_L0T2.R` now refuses to analyse silently-inconsistent inputs. It drops spreadsheet **totals rows** (rows with neither a culture code nor a taxon name — one such row was previously read as a genotype and plotted as a giant *Incertae sedis* category), verifies that pooled and sample-level isolate totals agree, and checks that culture codes are unique, since a repeated code would mark subsamples of one colony that must be collapsed before abundances mean anything. Finally, it compares every sampling unit with the curator’s sampling design: a mislabelled zone or unit does not raise an error, it silently becomes an extra “replicate”, which is how the Lauraceae came to be counted as 10 units instead of 8.

Scripts

Script	Purpose
<code>main.R</code>	Master script — loads libraries, sources all other scripts in order
<code>cluster_otus.sh</code>	Clusters the ITS sequences into OTUs (run once; needs <code>vsearch</code>)
<code>parse_L0T2.R</code>	Reads both workbooks, joins the OTU map, applies design rules, runs integrity checks
<code>plot_abundance.R</code>	Horizontal bar charts of isolate counts by taxonomic level
<code>plot_abundance_pooled.R</code>	Bar charts with uncertain taxa pooled as “ <i>Incertae sedis</i> ”
<code>plot_pie.R</code>	Pie charts of community composition (3 pies per level)
<code>community_analysis.R</code>	Multivariate community ecology; OTU-level tests plus multi-rank taxonomic sweeps (Section D)
<code>diversity_analysis.R</code>	Sampling completeness (Chao1/ACE/coverage) and replicated per-unit alpha diversity
<code>generate_readme.R</code>	Generates this README dynamically from analysis results

```
# From the project directory:  
source("main.R")
```

R packages used

Package	Version	Role
readxl	1.5.0.1	Reading Excel input
dplyr	1.2.1	Data manipulation
tidyr	1.3.2	Data reshaping
ggplot2	4.0.3	Plotting
scales	1.4.0	Axis formatting
vegan	2.7.5	Diversity, NMDS, PERMANOVA, ANOSIM, betadisper
indicspecies	1.8.0	Indicator species analysis (IndVal)
ggVennDiagram	1.5.7	Venn diagrams

Output Structure

```
plots/  
  abundance/                # Bar charts (raw)  
  abundance_pooled_incertae_sedis/ # Bar charts (uncertain taxa pooled)  
  pie_charts/              # Pie charts  
  community/              # Community ecology plots (NMDS, rarefaction, etc.)  
  png/                    # PNG versions for README embedding  
  
tables/                  # Statistical results and summary tables
```

Results

Throughout the Results, each analysis is introduced with a short **plain-language explanation** of what it measures and how to read it, followed by a **brief interpretation** of what the LOT2 data actually show. A synthesis of all findings is given in the final **Conclusion**.

A0. How much of the community was actually found?

Why this comes first. Every diversity number below is a *sample* statistic. Culture-based sampling of tropical endophytes recovers a fraction of what is present, and how large that fraction is determines what the rest of the section can claim. Two standard measures answer it: **Chao1/ACE** estimate the true richness from how many taxa were seen once or twice, and **Good's coverage** estimates the probability that the *next* isolate collected would belong to a taxon already seen.

Substrate	Units	Isolates	Observed	Chao1 ($\pm$ SE)	ACE	% of Chao1	Singletons	Coverage
Ficus leaves	10	264	74	135.8 $\pm$ 27.5	125.6	55%	39	85.2%
Ficus wood	12	167	64	159.7 $\pm$ 43.5	191.3	40%	42	74.9%
Lauraceae leaves	8	219	45	82.5 $\pm$ 20.6	90.9	55%	25	88.6%

Interpretation. Only **40-55% of the estimated richness was recovered**, with Good's coverage of 74.9-88.6%. Roughly one isolate in seven belongs to an OTU seen exactly once. Two things follow. First, **all observed richness values are substantial underestimates** and should be quoted alongside Chao1.

Second, *Ficus* wood is the least completely sampled substrate (40% of Chao1), so its apparently lower richness is partly a sampling artefact — which is why the comparison below is done per sampling unit and again at equal coverage.

Observed vs estimated richness

Unit: 97% ITS OTU | error bar = 1 SE of Chao1 | less than 55% of estimated richness was recovered

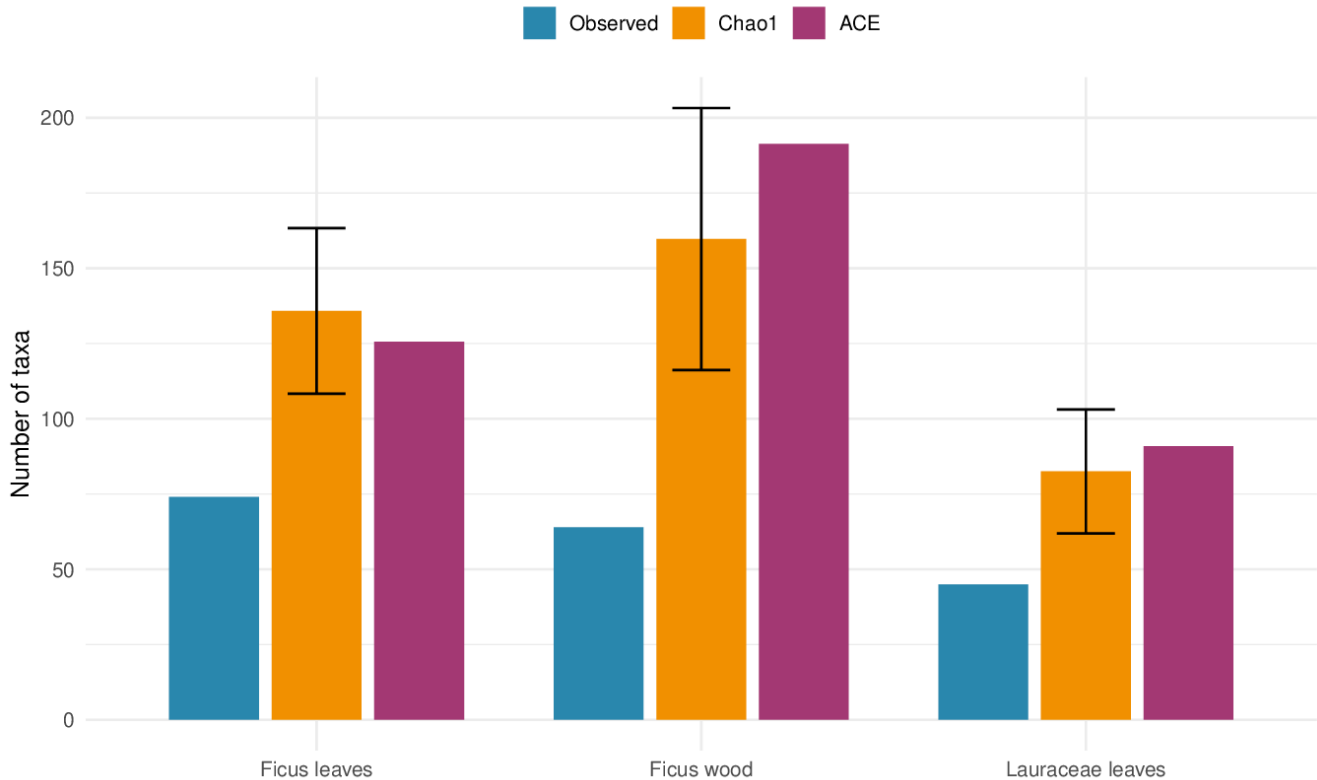


Figure 1: Observed vs estimated richness

Richness compared at equal coverage Comparing richness at equal *isolate count* still favours whichever substrate has the flatter abundance distribution. The fairer contrast standardises to equal **coverage** (Chao & Jost 2012) — how many taxa each substrate holds at the same level of sampling completeness.

Substrate	Isolates needed	Coverage reached	Richness at that coverage
Ficus leaves	108	74.6%	44.6
Ficus wood	155	74.2%	60.9
Lauraceae leaves	37	75.7%	16.1

Interpretation — and an apparent contradiction worth understanding. At equal coverage the ranking **reverses**: **Ficus wood** is the richest substrate, not the poorest. The two statements are not in conflict because they answer different questions. Per sampling unit (A1 below), a patch of wood carries **fewer** taxa than a patch of leaf — lower alpha diversity. But wood needs far more isolates to reach the same coverage, because its community has a much longer tail of rare taxa; pooled across the substrate it therefore holds **more** taxa overall. In short: **leaf units are individually richer, while the wood community as a whole is more diverse and more heterogeneous between units.** That heterogeneity is the same pattern the dispersion test picks up (Section D), and it is a substantive ecological result rather than an artefact.

Full tables: tables/sampling_completeness.csv, tables/richness_at_equal_coverage.csv.

A. Alpha Diversity

What it is. *Alpha diversity* describes how varied the fungal community is **within a single substrate**. It combines *richness* (how many taxa) with *evenness* (whether isolates are spread across taxa or dominated by a few).

Index	What it measures	Higher means
Richness (S)	Number of distinct taxa	More taxa present
Shannon (H')	Combines richness and evenness	More diverse
Hill q1 (= exp H')	Effective number of common taxa	More diverse
Simpson (1-D)	Probability two random individuals differ	More diverse
Pielou (J')	How evenly individuals are distributed	More even

A1. Per sampling unit — the version that can be tested

Diversity is computed **per sampling unit**, giving 10 / 12 / 8 independent values per substrate instead of one pooled number. That is what makes a significance test possible at all: a single pooled value per substrate has no variance and supports no inference.

Substrate	Units	Isolates	Mean S (±SD)	Mean H' (±SD)	Mean Hill q1	Mean J'
Ficus leaves	10	264	16.10 ± 2.38	2.595 ± 0.201	13.63	0.936
Ficus wood	12	167	8.42 ± 3.12	1.814 ± 0.518	6.82	0.884
Lauraceae leaves	8	219	12.88 ± 3.48	2.245 ± 0.298	9.83	0.889

Kruskal-Wallis across substrates (Holm-adjusted across the four indices):

Index	H	df	p	p (Holm)
richness_S	16.9596	2	0.0002076	0.0008304
shannon_H	15.7010	2	0.0003896	0.0011690
hill_q1	15.7010	2	0.0003896	0.0011690
pielou_J	4.8562	2	0.0882000	0.0882000

Pairwise follow-ups (Wilcoxon, Holm-adjusted):

Index	Comparison	p (Holm)
richness_S	Ficus wood vs Ficus leaves	0.0004728
richness_S	Lauraceae leaves vs Ficus leaves	0.0646900
richness_S	Lauraceae leaves vs Ficus wood	0.0288200
shannon_H	Ficus wood vs Ficus leaves	0.0009745
shannon_H	Lauraceae leaves vs Ficus leaves	0.0469400
shannon_H	Lauraceae leaves vs Ficus wood	0.0469400

Interpretation. **Richness and Shannon differ significantly between substrates** (Holm-adjusted p = 0.0008304 and 0.001169). *Ficus* leaves are the richest and most diverse, *Ficus* wood the poorest

(Holm $p = 0.0004728$ for richness), and the Lauraceae sits between them. The Lauraceae separates from *Ficus* wood (richness $p = 0.02882$, Shannon $p = 0.04694$) and only marginally from *Ficus* leaves (richness $p = 0.06469$, Shannon $p = 0.04694$). Before the sampling-unit corrections, two spurious one- and three-isolate Lauraceae units pulled its mean down and made it indistinguishable from wood. **Evenness (Pielou J') does not differ** ($p = 0.0882$): all three communities are similarly undominated, and the difference is in how many taxa are present, not how they are balanced. Because *Ficus* wood is also the least completely sampled substrate, part of its lower richness is sampling effort — the effect is real but its size should not be read off these means alone.

Per-sampling-unit richness_S

Each point is one sampling unit | Kruskal–Wallis Holm-adjusted $p = 0.0008304$ | unit: 97% ITS OTU

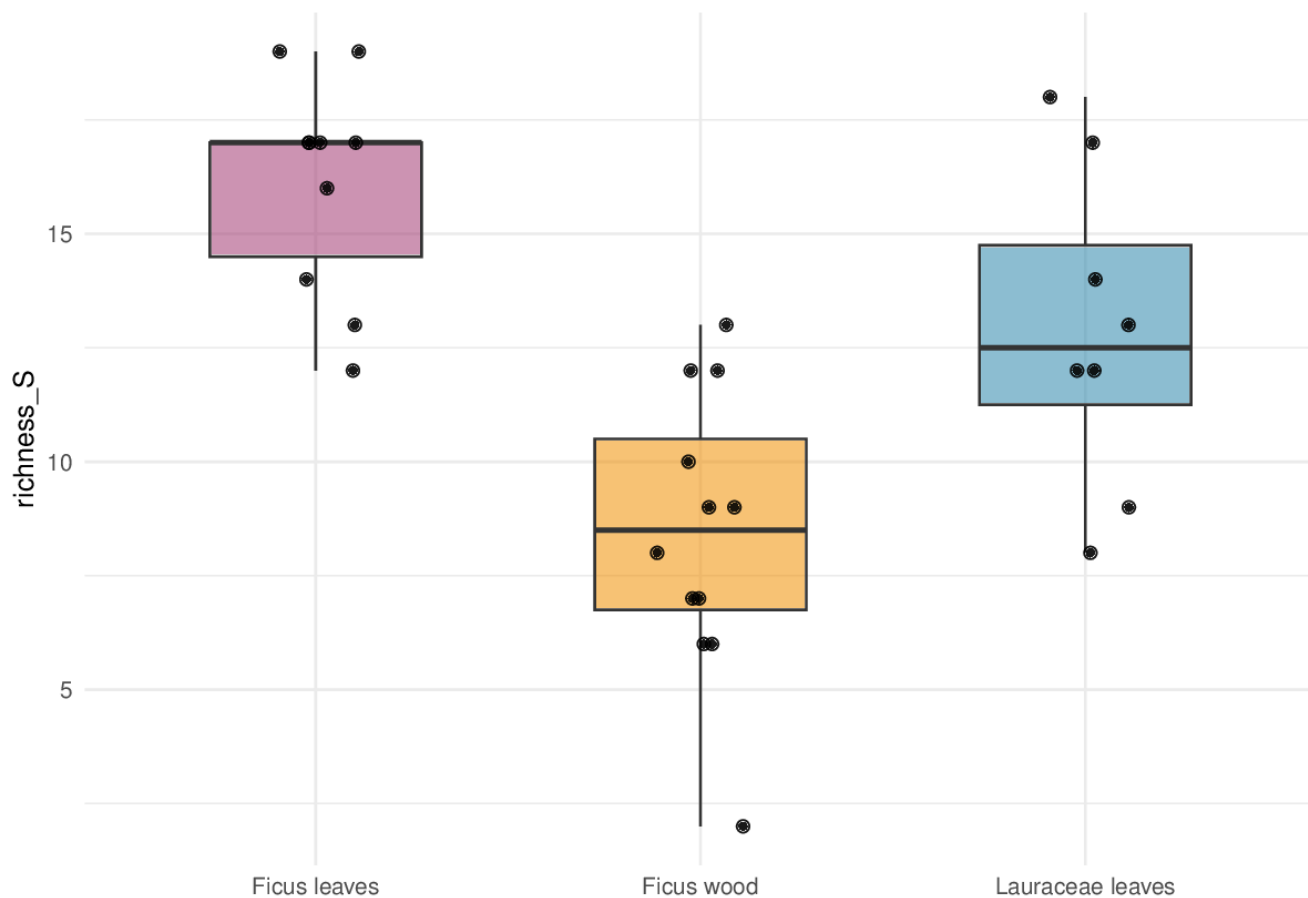


Figure 2: Richness per sampling unit

Per-unit values: tables/alpha_diversity_per_unit.csv; tests: tables/alpha_diversity_tests.csv, tables/alpha_diversity_pairwise.csv.

A2. Pooled indices across taxonomic ranks

The table below is the pooled view (one value per substrate per rank) retained for comparability with earlier versions of this report. **It carries no error and supports no test** — use A1 for inference. *Incertae sedis* taxa are excluded rank by rank.

At ITS taxon level

Per-sampling-unit shannon_H

Each point is one sampling unit | Kruskal-Wallis Holm-adjusted $p = 0.001169$ | unit: 97% ITS OTU

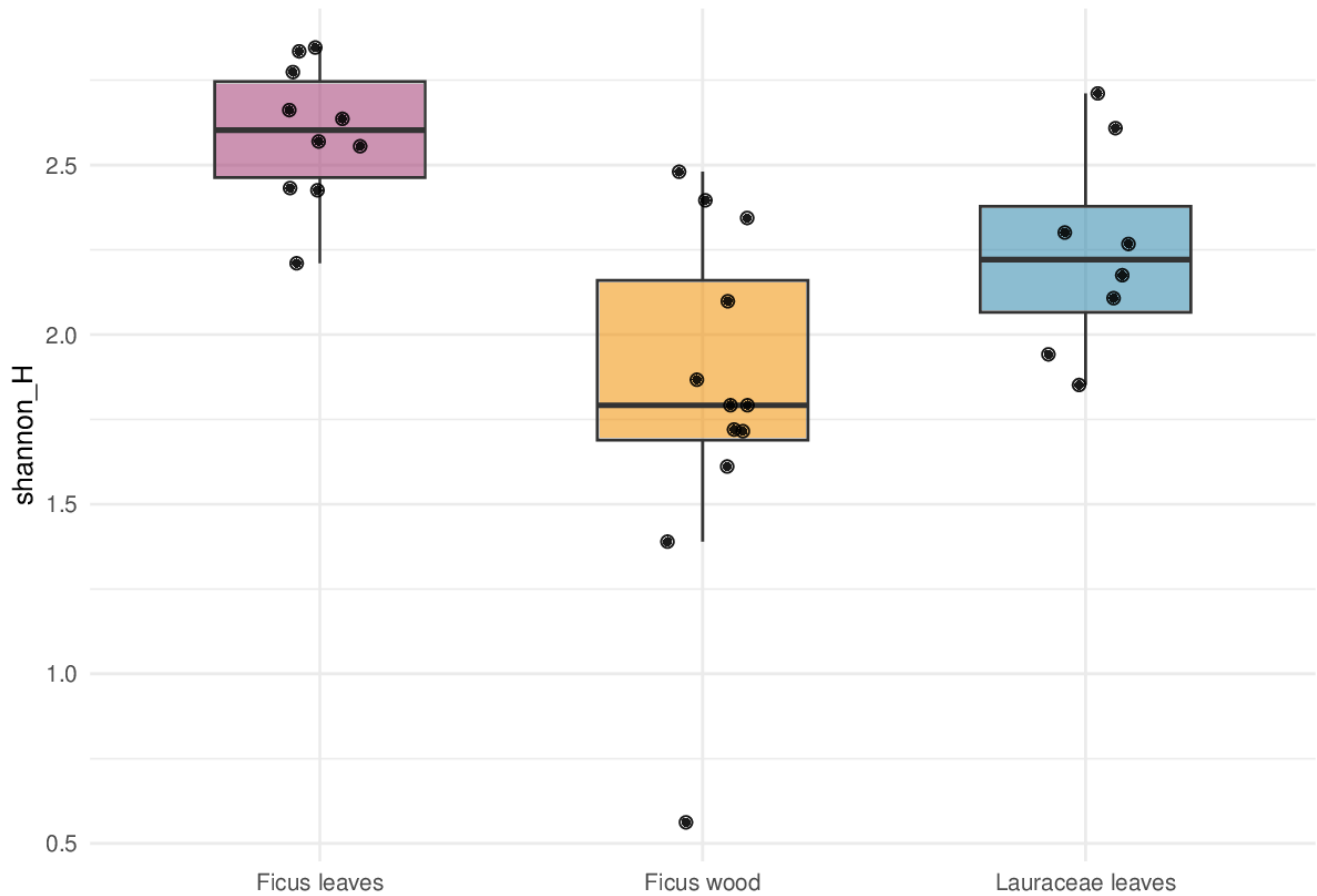


Figure 3: Shannon per sampling unit

Substrate	S	N	H'	1-D	Inv. Simp.	J'
Lauraceae leaves	106	219	4.2813	0.9781	45.7207	0.9181
Ficus leaves	135	264	4.4705	0.9791	47.8681	0.9114
Ficus wood	81	167	4.0073	0.9722	35.9858	0.9119

Interpretation. The richest community is **Ficus leaves** ($S = 135$ distinct ITS taxa), and the least rich is **Ficus wood** ($S = 81$). Shannon diversity is highest in **Ficus leaves** ($H' = 4.4705$) and lowest in **Ficus wood** ($H' = 4.0073$). Pielou evenness is high ($J' > 0.9$) for all three substrates, meaning no single genotype dominates any community — isolates are spread across many co-occurring taxa. Because richness partly reflects sampling effort (N differs between substrates), richness values should be compared together with the **rarefaction curves** below, which put all substrates on an equal-effort footing.

Diversity was also computed at every higher rank (phylum $\rightarrow$ genus); the full table is in `tables/alpha_diversity.csv`.

shannon_H across substrates

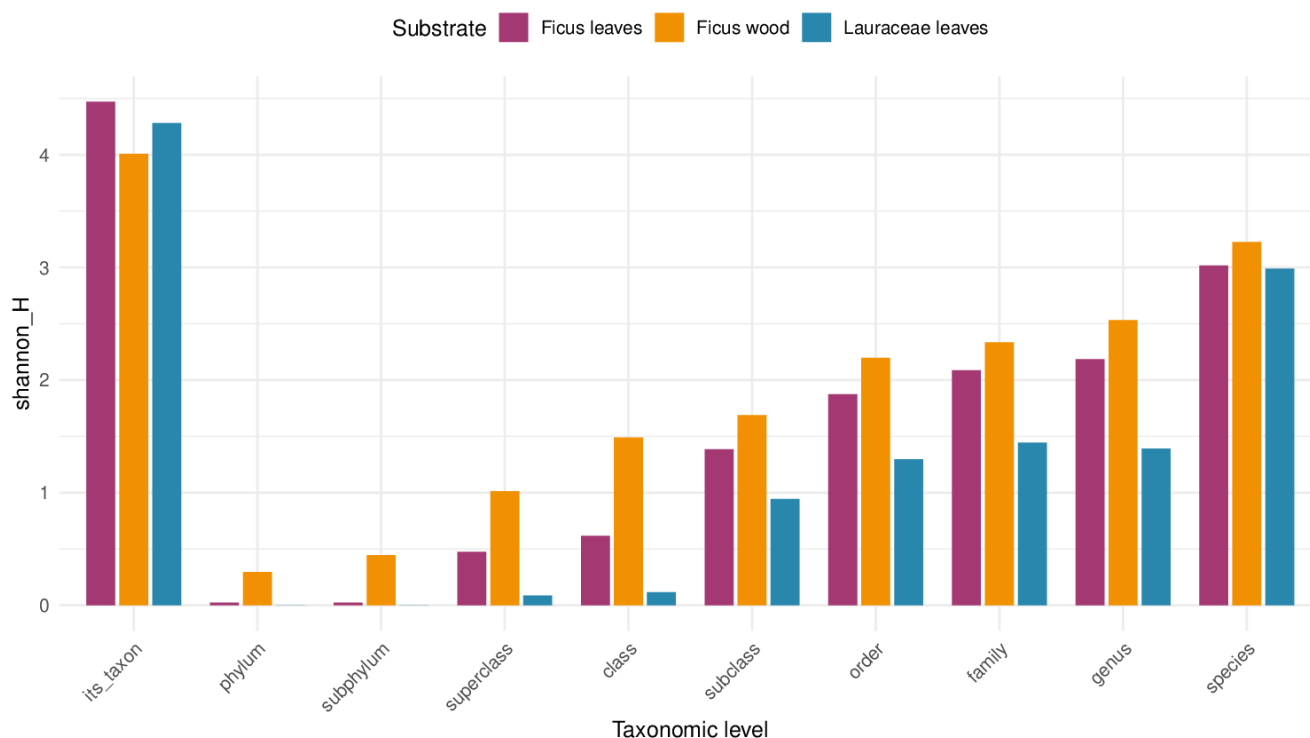


Figure 4: Shannon diversity

The bars show Shannon H' for each substrate across taxonomic ranks. H' naturally decreases towards coarser ranks (fewer categories), but the ranking of substrates stays broadly consistent, indicating the diversity differences are not an artefact of one particular rank.

B. Community Composition

These analyses describe **what the communities are made of** and **how much they overlap**. The rank-abundance curves and Venn diagrams exclude *Incertae sedis*; the relative-abundance bars show it, so that each bar is the full isolate census.

B1. Rank-Abundance Curves

What it is. Taxa are ranked from most to least abundant (x-axis) against their relative abundance on a log scale (y-axis). A **steep** curve means a few taxa dominate (low evenness); a **shallow, long** curve means many taxa share the community evenly (high evenness). The length of each curve reflects richness.

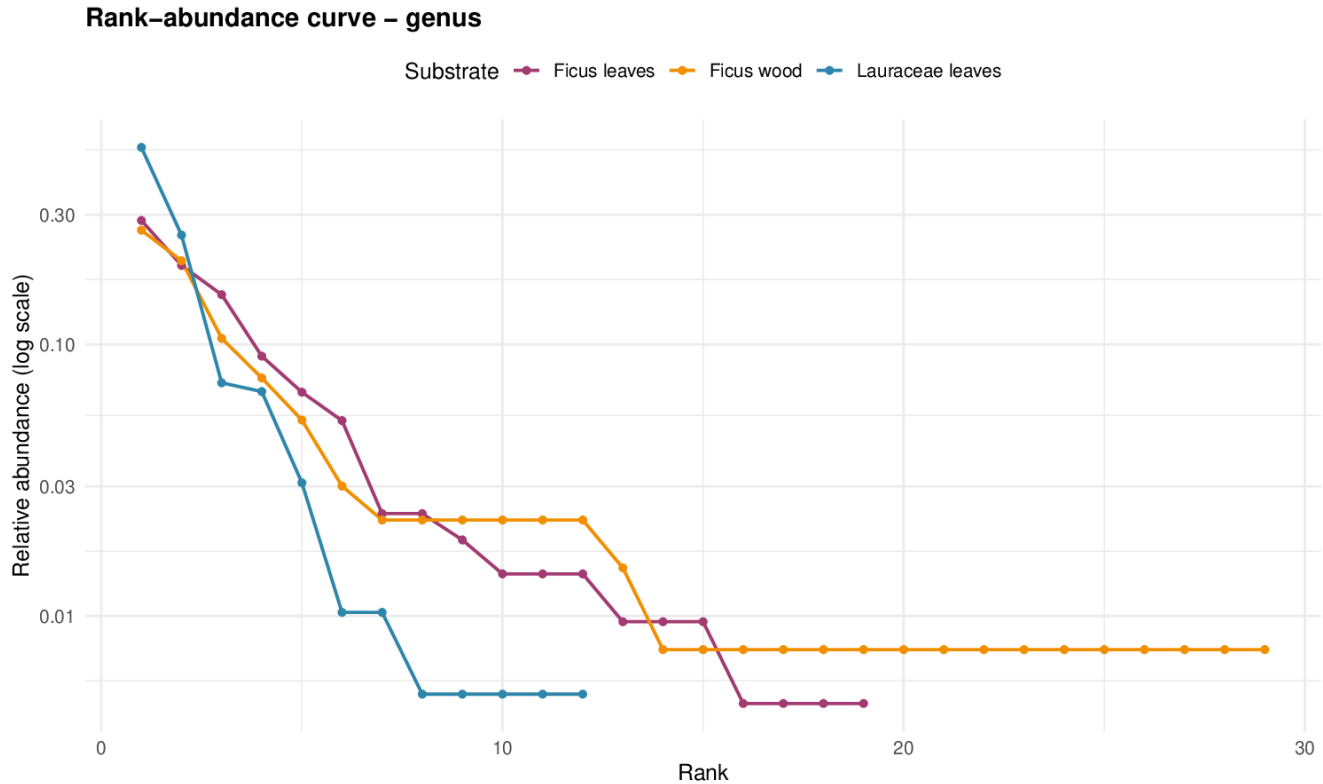


Figure 5: Rank-abundance at genus level

Interpretation. All three substrates show relatively shallow curves with long tails, confirming the high evenness seen in the Pielou index: communities are not dominated by one or two hyper-abundant genera but consist of many moderately frequent taxa plus a long tail of rare ones — a pattern typical of tropical endophyte assemblages.

B2. Relative Abundance

What it is. Stacked bars show the **proportional composition** of each substrate at a given rank. Each bar sums to 100% of **all** isolates: those that cannot be placed at that rank form the grey *Incertae sedis* segment on top, and its share is printed under each bar. They make it easy to see which phyla/genera dominate and how composition shifts between substrates.

Share of isolates that are *Incertae sedis* at each rank:

Rank	Lauraceae leaves	Ficus leaves	Ficus wood
phylum	0.0%	0.0%	1.8%
subphylum	0.0%	0.0%	0.0%
superclass	0.0%	0.4%	9.6%
class	0.0%	0.0%	0.6%
subclass	0.9%	9.5%	10.8%
order	0.0%	5.3%	2.4%
family	7.8%	12.1%	9.0%

Rank	Lauraceae leaves	Ficus leaves	Ficus wood
genus	11.4%	20.5%	20.4%
species	65.8%	53.8%	65.3%
its_taxon	0.0%	0.0%	0.0%

Interpretation. At phylum level the communities are overwhelmingly **Ascomycota**, as expected for culturable endophytes, and almost every isolate is placed (1.8% or less unresolved). The unresolved share grows towards the finer ranks, reaching up to 20.5% at genus and 65.8% at species level. The genus-level bars therefore name at least 79.5% of each substrate's isolates, and they reveal the real contrast between substrates: the identity and proportion of dominant genera differ markedly between leaves and wood, foreshadowing the significant substrate effect quantified in the multivariate tests below.

B3. Venn Diagrams — Shared Taxa

What it is. The Venn diagram counts how many taxa are **unique** to each substrate versus **shared** between them. It is a simple presence/absence view of community overlap (abundance is ignored).

Interpretation. At **genus level** (43 identified genera in total): **6** genera occur in all three substrates (a shared generalist core), while **2** are unique to Lauraceae leaves, **10** unique to Ficus leaves and **20** unique to Ficus wood. The substantial number of substrate-exclusive genera indicates a degree of **habitat specialisation** layered on top of a shared generalist core.

Multivariate Statistical Analyses

These analyses ask **whether whole communities differ between groups** (substrates, zones, positions). They use the sample-level data (**30 sampling units, 142 ITS genotypes**), the **Bray-Curtis dissimilarity** (a 0–1 measure of how different two samples are in both *which* taxa are present and *how abundant* they are), and **999 permutations** to obtain p-values without assuming normality.

How to read each test:

- **NMDS ordination** — squeezes the many-dimensional Bray-Curtis distances into a 2-D map so that samples plotting close together have similar communities. The **stress** value measures distortion: < 0.10 excellent, < 0.20 acceptable, > 0.20 unreliable. Crosses mark group centroids; shaded ellipses show 95% confidence regions. **A stress at or near zero is a warning, not a triumph** — see the caveat below, which explains why these plots report two stress values.
- **PERMANOVA** (adonis2) — tests whether **group centroids differ**. **R²** is the fraction of community variation explained by the grouping (effect size); a small **p** means the separation is unlikely by chance.
- **ANOSIM** — a complementary rank-based test; **R** ranges from 0 (no separation) to 1 (groups completely distinct).
- **Beta-dispersion / PERMDISP** (betadisper) — checks whether groups differ in **within-group spread** rather than location. If PERMDISP is significant, part of a PERMANOVA result may reflect unequal dispersion rather than a pure shift in composition, so it is an important caveat.
- **Pairwise PERMANOVA** — which specific pairs of groups differ, with Holm correction for multiple tests.
- **Rarefaction** — expected richness rescaled to equal sampling effort, so richness can be compared fairly.
- **Indicator species (IndVal)** — identifies taxa statistically associated with (diagnostic of) a particular group. One test is run **per taxon**, so the raw p-values are corrected with Benjamini-Hochberg and only taxa passing **FDR q ≤ 0.05** are reported as indicators. Without that correction, ~14 of ~280 taxa pass p ≤ 0.05 by chance alone.

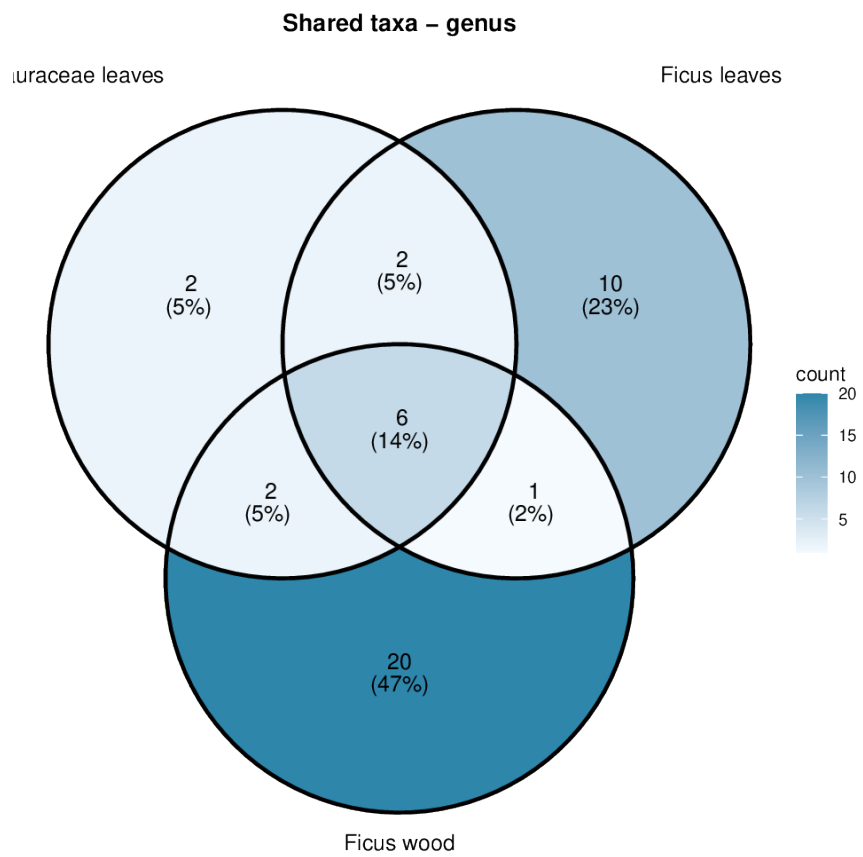


Figure 8: Venn diagram — genus

An important caveat: why the ITS-level ordinations collapse onto a line

Some NMDS plots in this report (orientation at its_taxon level) show most samples squeezed onto a single near-vertical line with one point flung far away, next to a stress of ~ 0.0001 . That is **not** an excellent fit and **not** a plotting bug — it is a known failure mode of NMDS, and it is worth understanding because it determines which figures can be read.

The cause. At ITS-genotype resolution this dataset is dominated by rare taxa: **52% of the 142 genotypes were isolated exactly once**, so **37% of all sampling-unit pairs share no genotype whatsoever**. Every one of those pairs has a Bray-Curtis dissimilarity of *exactly* 1 — they are **tied**.

NMDS fits an ordination by rank order, and by default (monoMDS, weak/primary ties) **tied dissimilarities are allowed to map to any distances at all**. With well over half of the pairs tied, the optimiser is therefore free to ignore most of the matrix: it only has to get the *ordering* of the minority of pairs that do share taxa right. It can do that almost perfectly in two dimensions — hence the near-zero stress — while pushing the unconstrained samples wherever is convenient. The collapsed line and the distant outlier are those unconstrained samples.

The evidence. Refitting the same dissimilarities with *strong* (secondary) ties, which force tied pairs to equal distances, gives the honest answer. At ITS level the stress rises from **0.0882 to 0.2031** — i.e. *unreliable* by the usual thresholds. At genus level the two figures agree closely (0.1617 vs 0.1128), confirming that those ordinations are real. Every NMDS plot in this report now prints **both** values, and is labelled UNRELIABLE on any of three grounds: a tie-aware stress above 0.20, 9 or fewer sampling units (at or below $4k+1$ points a 2-D solution fits almost anything), or a tie-aware stress below 0.001 (a perfect fit means the configuration is unconstrained, not faithful).

Two further points were checked and ruled out as explanations: metaMDS applied **no** data transformation here, and it did **not** fall back to extended (step-across) dissimilarities — the largest dissimilarity fed to the ordination is exactly 1. The analysis is doing what it says; the data simply cannot support a 2-D map at genotype resolution.

The dissimilarity matrix is nonetheless **connected**: every sampling unit is linked to the others through shared 97% ITS OTUs, so any collapse comes from the ties alone, not from isolated samples. (Earlier versions of this report named a disconnected one-isolate unit, Lauraceae leaves __Z5__S4. That was true only for the name-based genotypes, and the unit itself was an artefact of the zone-labelling error. It is now part of branch S4.)

What to do with this:

- **Do not read the ITS-level NMDS maps.** They are kept because they are referenced throughout the literature-standard workflow, and each is now labelled.
- **PERMANOVA, ANOSIM and IndVal are unaffected.** They operate on the dissimilarity matrix itself — its sums of squares and rank order — never on a 2-D embedding, so no tie-handling choice enters. These are the results to trust.
- **Use the higher-rank ordinations in Section D as the readable maps.** Grouping isolates into genera or families dissolves the singleton problem: the share of pairs sharing no taxa falls from 51% at ITS level to 29% at genus level, and the substrate signal is reproduced there with an honest stress.

C1. Substrate Comparison: Ficus Leaves vs Ficus Wood vs Lauraceae Leaves

This is the **headline comparison**: do the three substrates host different fungal communities?

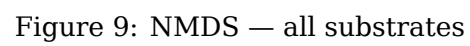
NMDS Ordination

Stress = 0.1376 (acceptable to good — the 2-D map is a faithful summary).

Interpretation. The three substrates form visually distinct clouds, with the two leaf substrates sitting closer to each other than to wood — consistent with tissue type (leaf vs wood) being a strong driver

Bray–Curtis | stress 0.138 (weak ties) / 0.1582 (tie-aware) | 37% of sample pairs share no taxa

substrate  Ficus leaves  Lauraceae leaves
 Ficus wood



NMDS with top 15 taxa – substrate_all

Bray–Curtis | stress 0.138 (weak ties) / 0.1582 (tie-aware) | 37% of sample pairs share no taxa



Figure 10: NMDS with species overlay

(Section D1 confirms this at every taxonomic rank). The species overlay points to the genotypes pulling each substrate apart.

PERMANOVA

PERMANOVA – Bray-Curtis distance

Formula: community ~ substrate

Permutations: 999

Permutation test for adonis under reduced model

Permutation: free

Number of permutations: 999

```
adonis2(formula = form, data = meta, permutations = 999, method = "bray")
```

```
      Df SumOfSqs      R2      F Pr(>F)
Model    2   3.6803 0.33338 6.7516  0.001 ***
Residual 27   7.3588 0.66662
Total   29  11.0391 1.00000
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

F = 6.7516, R² = 0.33338, p = 0.001 — the substrate effect is **statistically significant**. Substrate explains about 33% of the total community variation, a large effect for field endophyte data.

Adjusted for sampling depth Sampling units differ about two-fold in how many isolates they yielded (*Ficus* wood units gave roughly half as many as leaf units), and depth on its own predicts composition. Re-fitting with depth as a covariate (marginal / type-III): **substrate R² = 0.2107, p = 0.001; depth R² = 0.0406, p = 0.054**. The substrate effect therefore survives adjustment — it is not an artefact of unequal recovery. Once substrate is accounted for, depth adds nothing significant of its own. Details in tables/substrate_all_permanova.txt.

ANOSIM

R = 0.5816, p = 0.001 — **statistically significant**. An R of this magnitude confirms that between-substrate differences clearly exceed within-substrate variation.

Beta-dispersion (PERMDISP)

F = 21.658, p = 0.001 — dispersion differences are **statistically significant**. Because within-group spread also differs, the PERMANOVA result partly reflects unequal dispersion; the significant separation on the NMDS nonetheless supports a genuine compositional shift.

Pairwise PERMANOVA

Which substrate *pairs* differ (Holm-corrected p). *** p<0.001, ** p<0.01, * p<0.05.

Comparison	F	R ²	p (raw)	p (Holm)	Sig.
Ficus leaves vs Ficus wood	6.8597	0.2554	0.001	0.003	**
Ficus leaves vs Lauraceae leaves	5.7062	0.2629	0.001	0.003	**
Ficus wood vs Lauraceae leaves	7.2505	0.2871	0.001	0.003	**

Interpretation. Every pair of substrates differs significantly after correction — each substrate carries a distinguishable community, not merely one odd substrate against two similar ones.

Rarefaction

Expected number of taxa if every substrate had been sampled to the same number of isolates — a fair richness comparison that removes the effect of unequal sampling effort.

Rarefaction curves – substrate_all

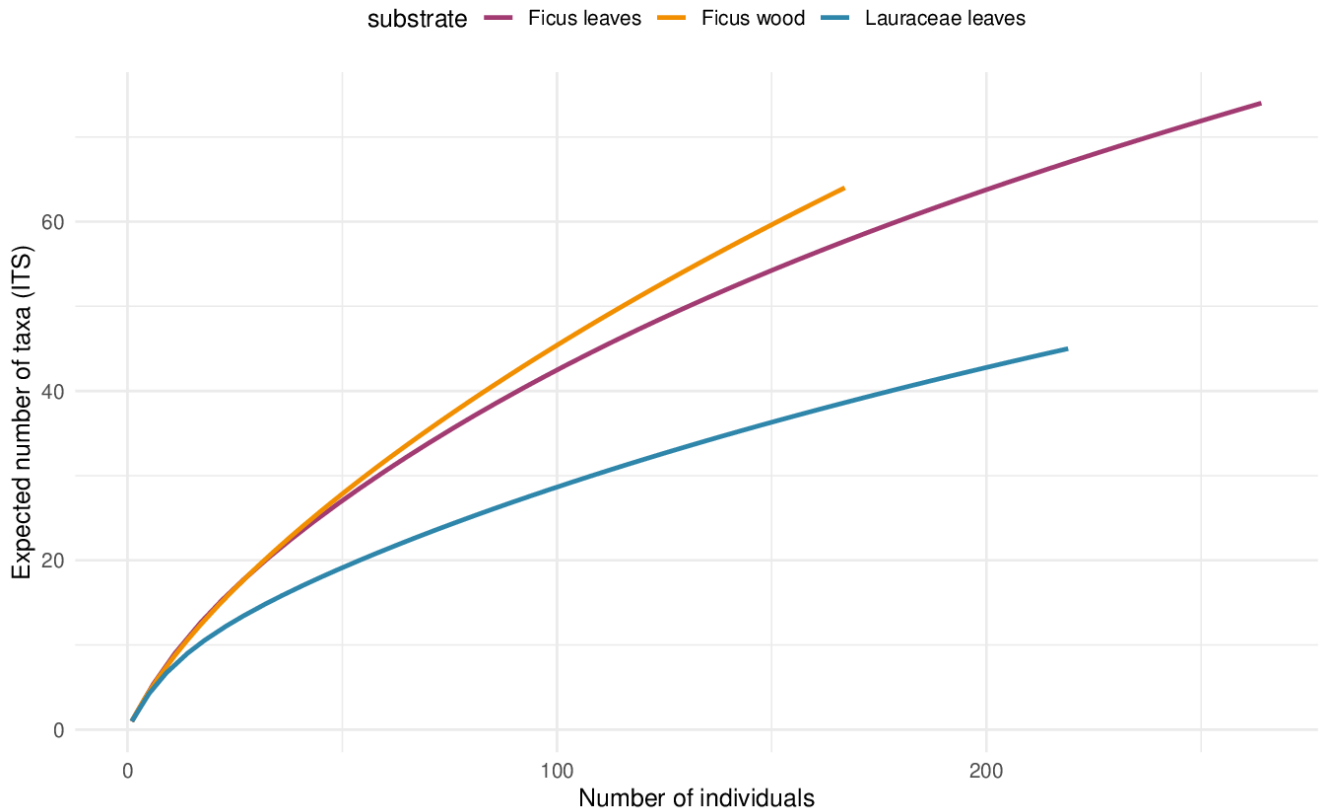


Figure 11: Rarefaction — all substrates

Interpretation. None of the curves has fully levelled off, so additional sampling would still recover new taxa in every substrate (the communities are undersampled, as usual for hyper-diverse tropical fungi). The **relative ordering** of the curves indicates which substrate is richest at equal effort, which is the sampling-fair complement to the raw richness values in Section A.

Indicator Species

Indicator (IndVal) analysis looks for taxa that are **diagnostic** of a substrate — both faithful to it and frequent within it. Each taxon is a separate test, so p-values are corrected across the whole taxon set. **21 taxa reach raw $p \leq 0.05$, but 13 survive FDR correction ($q \leq 0.05$).**

Interpretation. These taxa survive correction and can be treated as genuine markers. Full ranked list: `tables/indicator_species_significant.csv`; all taxa with p- and q-values: `tables/substrate_all_indicator_species_all.csv`.

C2. Ficus Leaves vs Lauraceae Leaves (leaf substrates only)

Why. Both are **leaf** endophyte communities but from different host plants, so this isolates the **host effect** from the leaf-vs-wood tissue effect.

PERMANOVA: $F = 5.7062$, $R^2 = 0.26288$, $p = 0.001$ — statistically significant. Even between two leaf communities, host identity (Ficus vs Lauraceae) leaves a detectable signature, explaining about 26% of the variation.

NMDS – substrate_leaves

Bray–Curtis | stress 0.141 (weak ties) / 0.1471 (tie-aware) | 0% of sample pairs share no taxa

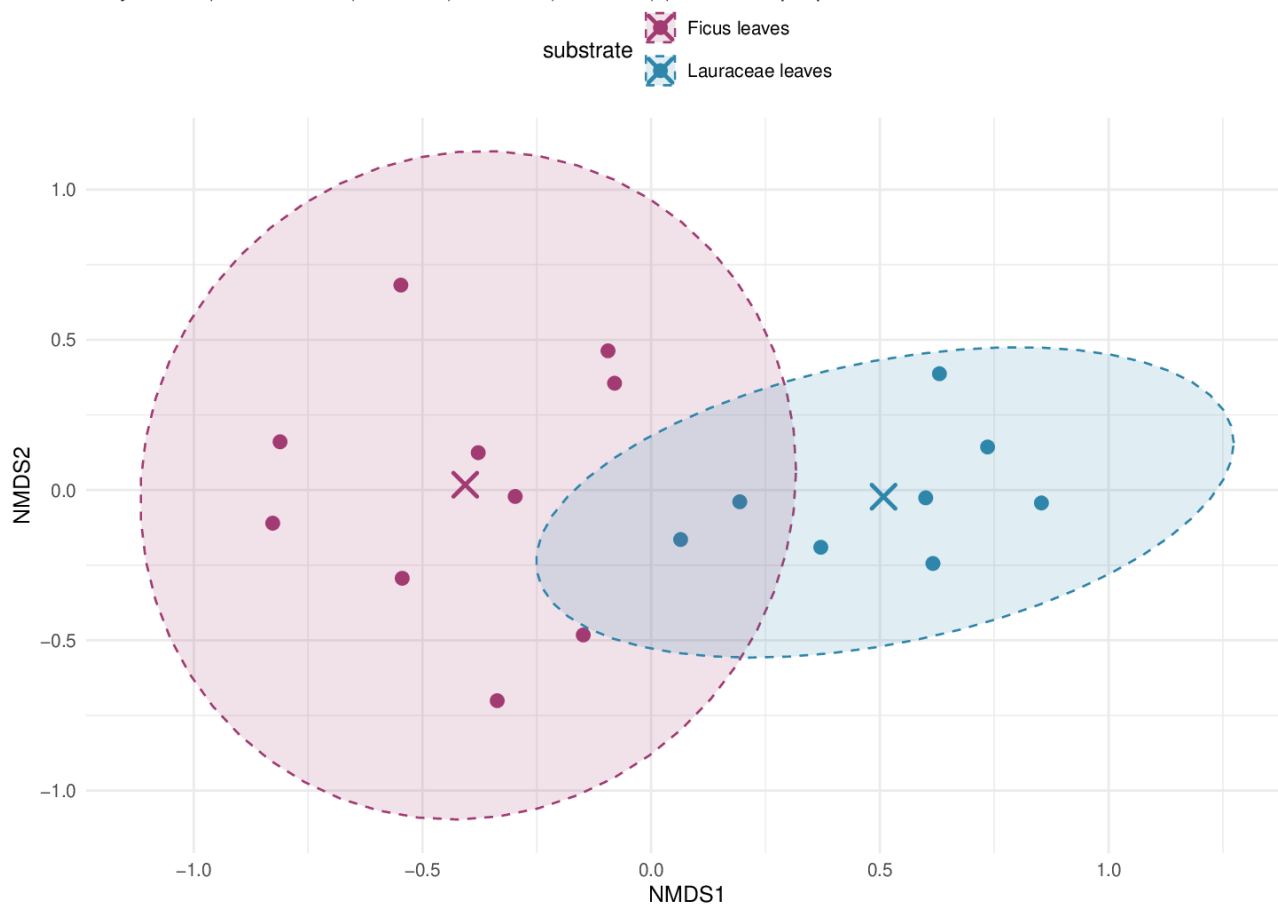


Figure 12: NMDS — Ficus vs Lauraceae leaves

Full results in tables/substrate_leaves_*.txt

D. Multi-level Taxonomic Analyses

Why this section exists. The multivariate tests above use **ITS genotypes**, the finest possible resolution. But an isolate that is *Incertae sedis* at species level often still has a defined **genus, family or order**. By linking every isolate to its **full lineage** (from the pooled taxonomy) we can repeat the community comparisons at each rank and ask: **is the substrate/zone signal a fine-scale artefact, or does it hold when isolates are grouped into higher, more confidently identified taxa?**

Lineage coverage — isolates with a defined value at each rank (out of 650):

Rank	Isolates resolved
phylum	643 / 650
class	645 / 650
order	628 / 650

Rank	Isolates resolved
family	585 / 650
genus	540 / 650
species	384 / 650
otu_97	650 / 650
its_taxon	650 / 650

D1. Substrate comparison across ranks

Rank	Taxa	No-taxa pairs	Stress	Tie-aware	F	R ²	p	ANOSIM R	ANOSIM p	PERMDISP p
phylum	4	0%	0.0274	0.0289	17.7325	0.5678	0.001	0.3332	0.001	0.001
class	9	0%	0.1416	0.0943	17.9676	0.5710	0.001	0.4679	0.001	0.001
order	25	7%	0.1500	0.1088	16.6664	0.5525	0.001	0.6584	0.001	0.001
family	34	17%	0.1467	0.1071	13.9714	0.5086	0.001	0.6548	0.001	0.001
genus	43	29%	0.1617	0.1128	12.6904	0.4845	0.001	0.6512	0.001	0.001
species	74	44%	0.0879	0.1480	7.1880	0.3474	0.001	0.5208	0.001	0.001
otu_97	142	37%	0.1376	0.1582	6.7516	0.3334	0.001	0.5816	0.001	0.001
its_taxon	284	51%	0.0882	0.2031	3.9544	0.2266	0.001	0.6713	0.001	0.002

F / *R*² / *p* are PERMANOVA; “No-taxa pairs” is the share of sample pairs with no taxon in common; “Tie-aware” is the stress refitted with strong ties — the honest one to judge by.

Interpretation. The substrate effect is **significant at every taxonomic rank** (PERMANOVA $p \leq 0.004$). Crucially, the effect size does **not** weaken when isolates are grouped into higher taxa — it is *strongest* around **class level** ($R^2 \approx 0.57$), compared with $R^2 \approx 0.23$ at ITS level. In other words, the substrates differ not just in which fine genotypes they carry, but in their broad taxonomic make-up, and removing the *Incertae sedis* noise sharpens rather than blurs that separation.

Which substrates are most alike? The overall test says the three substrates differ; it does not say which two are most alike. That is a question about where each community sits, so it is answered with the distance between substrate centroids, repeated at every rank:

Rank	Ficus leaves vs Ficus wood	Ficus leaves vs Lauraceae	Ficus wood vs Lauraceae
phylum	0.3516	0.022	0.372
class	0.5605	0.1876	0.6555
order	0.7367	0.3684	0.7441
family	0.7366	0.3937	0.7485
genus	0.7202	0.427	0.7723
species	0.6558	0.4365	0.6805
otu_97	0.6263	0.4939	0.6757
its_taxon	0.5308	0.5207	0.5759

Bray-Curtis distance between substrate centroids (PCoA space); the smallest per rank, i.e. the most similar pair, is in bold. Every pair also differs by PERMANOVA ($p \leq 0.001$) except Ficus leaves vs Lauraceae leaves at phylum level ($p = 0.344$).

Interpretation. At every rank the two leaf substrates have the closest centroids: **tissue type (leaf vs wood) is the strongest split**, and the host contrast between the two leaf communities is the smaller one. Pairwise PERMANOVA R^2 (also in the table file) is not used to rank the pairs, because R^2 falls when a group is internally heterogeneous, and *Ficus* wood units are far more heterogeneous than

NMDS at genus level – substrate_bylevel

Bray–Curtis | stress 0.1617 (weak ties) / 0.1128 (tie-aware) | 43 taxa | 29% of sample pairs share no taxa

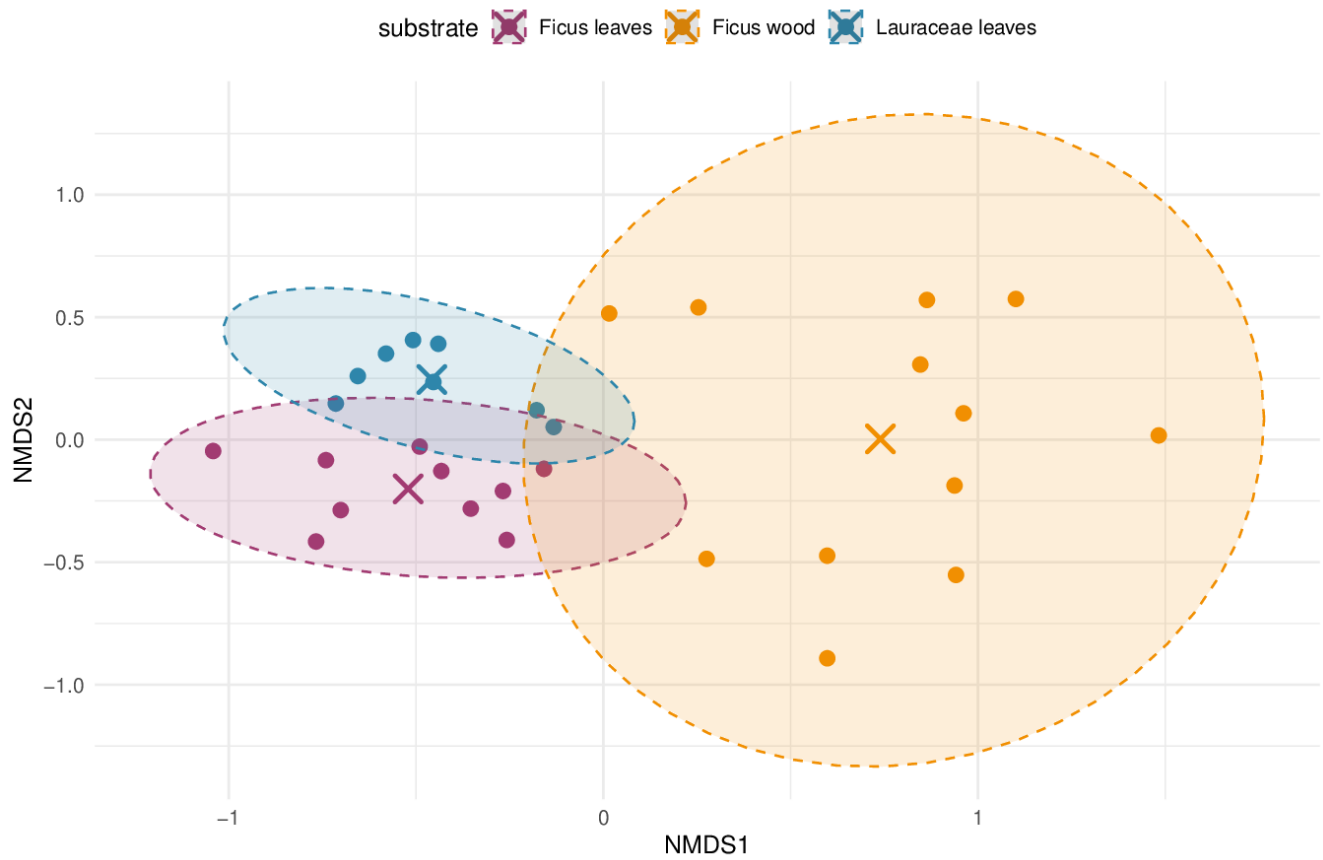


Figure 13: NMDS at genus level — substrates

NMDS at family level – substrate_bylevel

Bray–Curtis | stress 0.1467 (weak ties) / 0.1071 (tie-aware) | 34 taxa | 17% of sample pairs share no taxa

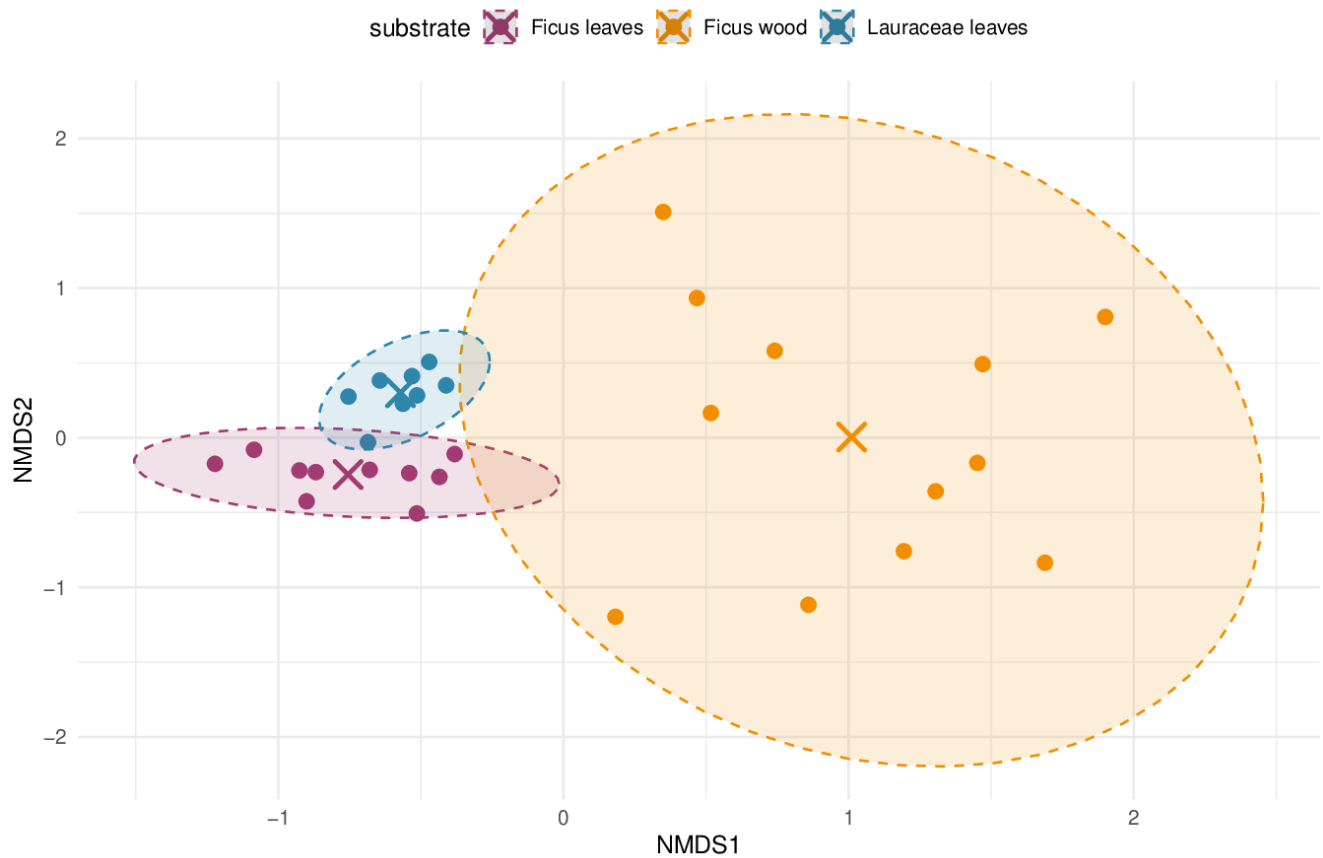


Figure 14: NMDS at family level — substrates

leaf units (PERMDISP, Section C1). That penalises every contrast involving wood regardless of how far apart the communities sit.

Full table: tables/substrate_pairwise_multilevel.csv.

D2. Ficus wood zones across ranks

Rank	Taxa	No-taxa pairs	Stress	Tie-aware	F	R ²	p	ANOSIM R	ANOSIM p	PERMDISP p
phylum	4	0%	0.0415	0.0537	0.5472	0.3132	0.839	- 0.2042	0.796	—
class	9	0%	0.1411	0.0534	1.7008	0.5863	0.017	0.3852	0.009	—
order	19	2%	0.1280	0.0929	1.1742	0.4946	0.231	0.2085	0.120	—
family	26	3%	0.1332	0.1312	1.0087	0.4567	0.471	0.0497	0.422	—
genus	29	6%	0.1456	0.3473	0.8914	0.4262	0.732	- 0.0825	0.671	—
species	30	42%	0.0906	0.1932	0.8047	0.4014	0.967	- 0.2825	0.970	—
otu_97	64	29%	0.1226	0.1683	0.8452	0.4133	0.848	- 0.2317	0.834	—
its_taxon	82	39%	0.0859	0.1955	0.9401	0.4393	0.707	- 0.1175	0.711	—

F / R² / p are PERMANOVA; “No-taxa pairs” is the share of sample pairs with no taxon in common; “Tie-aware” is the stress refitted with strong ties — the honest one to judge by.

Interpretation. Zones differ significantly only at **class** level ($p = 0.017$); at the other ranks $p \geq 0.231$. A significant result at 1 of 8 ranks, with no correction across ranks and no replication within any trunk zone (see E1), is weak evidence of a height effect.

D3. Substrate × position across ranks

Rank	Taxa	No-taxa pairs	Stress	Tie-aware	F	R ²	p	ANOSIM R	ANOSIM p	PERMDISP p
phylum	4	0%	0.0274	0.0289	13.1125	0.6021	0.001	0.4625	0.001	0.001
class	9	0%	0.1416	0.0943	23.9356	0.7342	0.001	0.6933	0.001	0.001
order	25	7%	0.1500	0.1088	16.3404	0.6534	0.001	0.7894	0.001	0.011
family	34	17%	0.1467	0.1071	13.3685	0.6067	0.001	0.7875	0.001	0.001
genus	43	29%	0.1617	0.1128	11.6545	0.5735	0.001	0.7788	0.001	0.001
species	74	44%	0.0879	0.1480	6.2131	0.4176	0.001	0.6721	0.001	0.001
otu_97	142	37%	0.1376	0.1582	5.9666	0.4077	0.001	0.7277	0.001	0.002
its_taxon	284	51%	0.0882	0.2031	3.4886	0.2870	0.001	0.7864	0.001	0.046

F / R² / p are PERMANOVA; “No-taxa pairs” is the share of sample pairs with no taxon in common; “Tie-aware” is the stress refitted with strong ties — the honest one to judge by.

Interpretation. Combining substrate with trunk/branch position remains significant at every rank, and the effect size again peaks at **class** level. This mirrors the substrate result: the signal is carried by broad taxonomic groups, not just rare fine-scale genotypes.

Summary tables: tables/substrate_multilevel_summary.csv, tables/ficus_wood_zones_multilevel_summary, tables/substrate_x_position_multilevel_summary.csv.

E. Zone-Based Analyses

These test whether **height on the tree** (trunk zones 1–5 vs canopy branch zone 6) structures the community. All are run at ITS-genotype level; Section D2 already showed the zone question is also answered the same way at higher ranks.

E1. Ficus Wood Across Zones

Community comparison of Ficus wood isolates collected at different tree zones (1–6).

Replication. Each trunk zone (1–5) is a **single sampling unit** (S11–S15); only zone 6 is replicated (7 branch units). A six-level zone factor therefore spends five degrees of freedom on five unreplicated trunk units, and its residual variation comes entirely from the zone-6 branches. This test cannot tell zones apart from units. It asks only whether the trunk units, taken one by one, depart from the spread among branches. **E2 is the replicated version of the height question.**

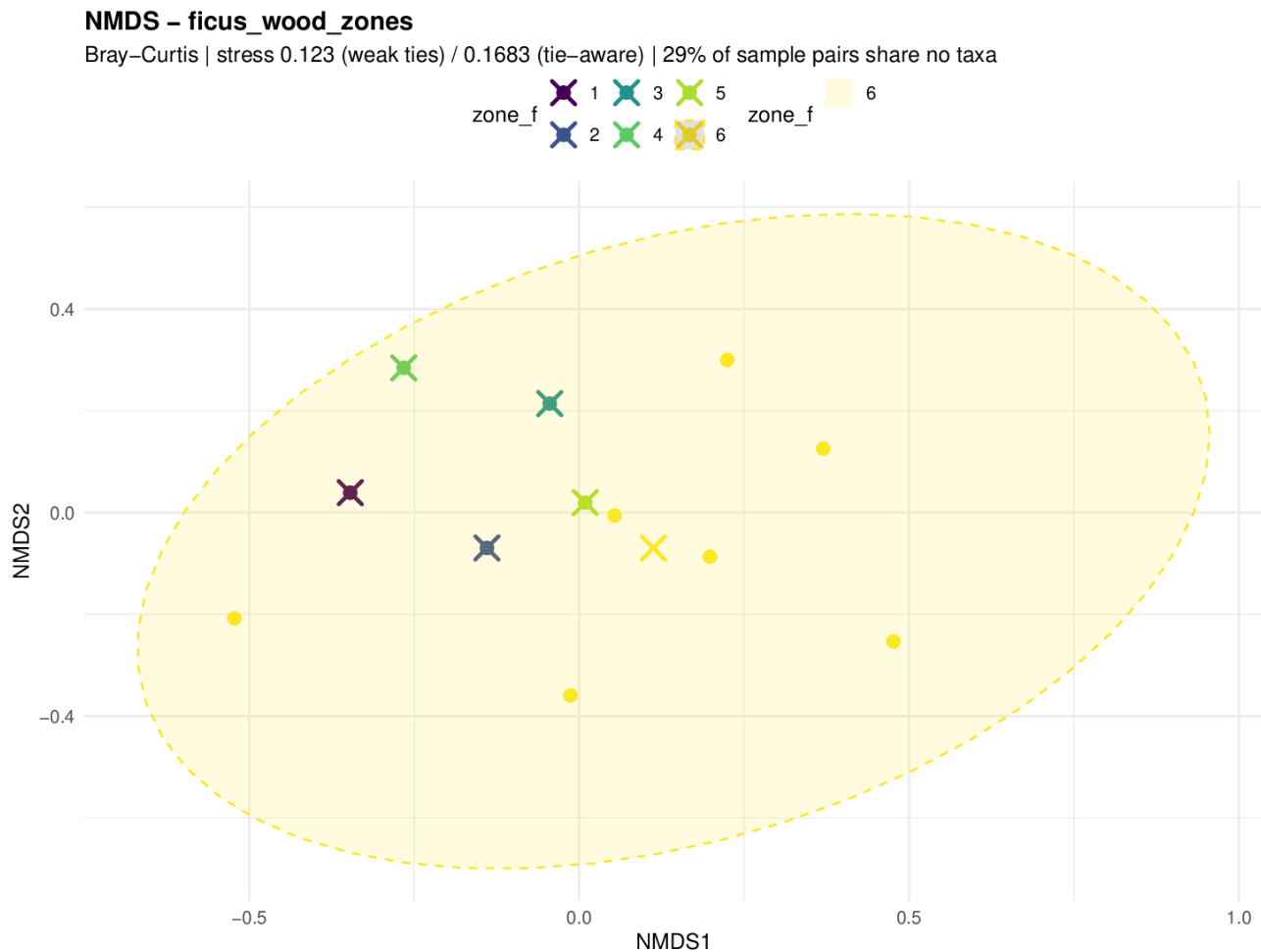


Figure 15: NMDS — Ficus wood zones

Interpretation. Samples from different zones intermingle on the NMDS with no zone-wise grouping. Given the replication above, that is consistent with height not mattering, but it is not evidence of it. Full results: tables/ficus_wood_zones_*.txt.

E2. Ficus Wood: Trunk (Zones 1-5) vs Branch (Zone 6)

PERMANOVA: $F = 2.3464$, $R^2 = 0.19005$, $p = 0.003$ — statistically significant. Trunk and branch wood host detectably different communities.

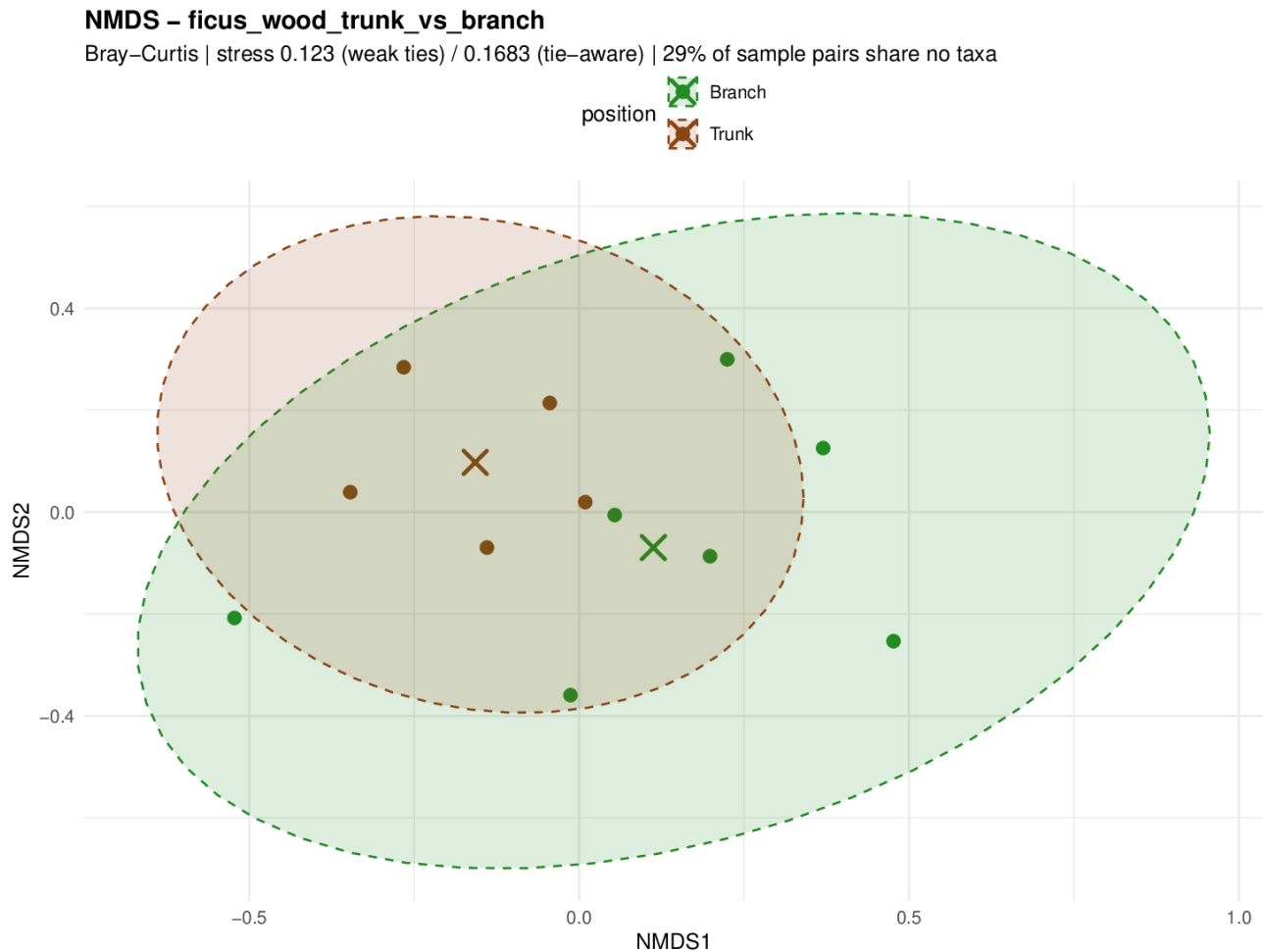


Figure 16: NMDS — trunk vs branch

E4. Substrate Comparison at Zone 6 (Branch Level Only)

Why. Restricting to zone 6 removes any confound between substrate and height, since all three substrates are present there. Now that every Lauraceae branch is correctly placed in zone 6, this comparison uses all 10 *Ficus* leaf units, all 8 Lauraceae units and the 7 *Ficus* branch-wood units. It is the full dataset minus the *Ficus* trunk.

PERMANOVA: $F = 5.4193$, $R^2 = 0.33006$, $p = 0.001$ — statistically significant. The substrate effect persists even within a single zone, confirming it is driven by substrate itself and not by differences in sampling height.

E5. Substrate × Position Interaction

Combined-factor analysis testing whether communities differ across substrate-position combinations (e.g. *Ficus* wood - Trunk vs *Ficus* wood - Branch vs *Ficus* leaves - Branch ...).

Interpretation. Groups separate primarily **by substrate**, with position adding only minor structure — substrate is the dominant organiser of these endophyte communities (see also the multi-rank

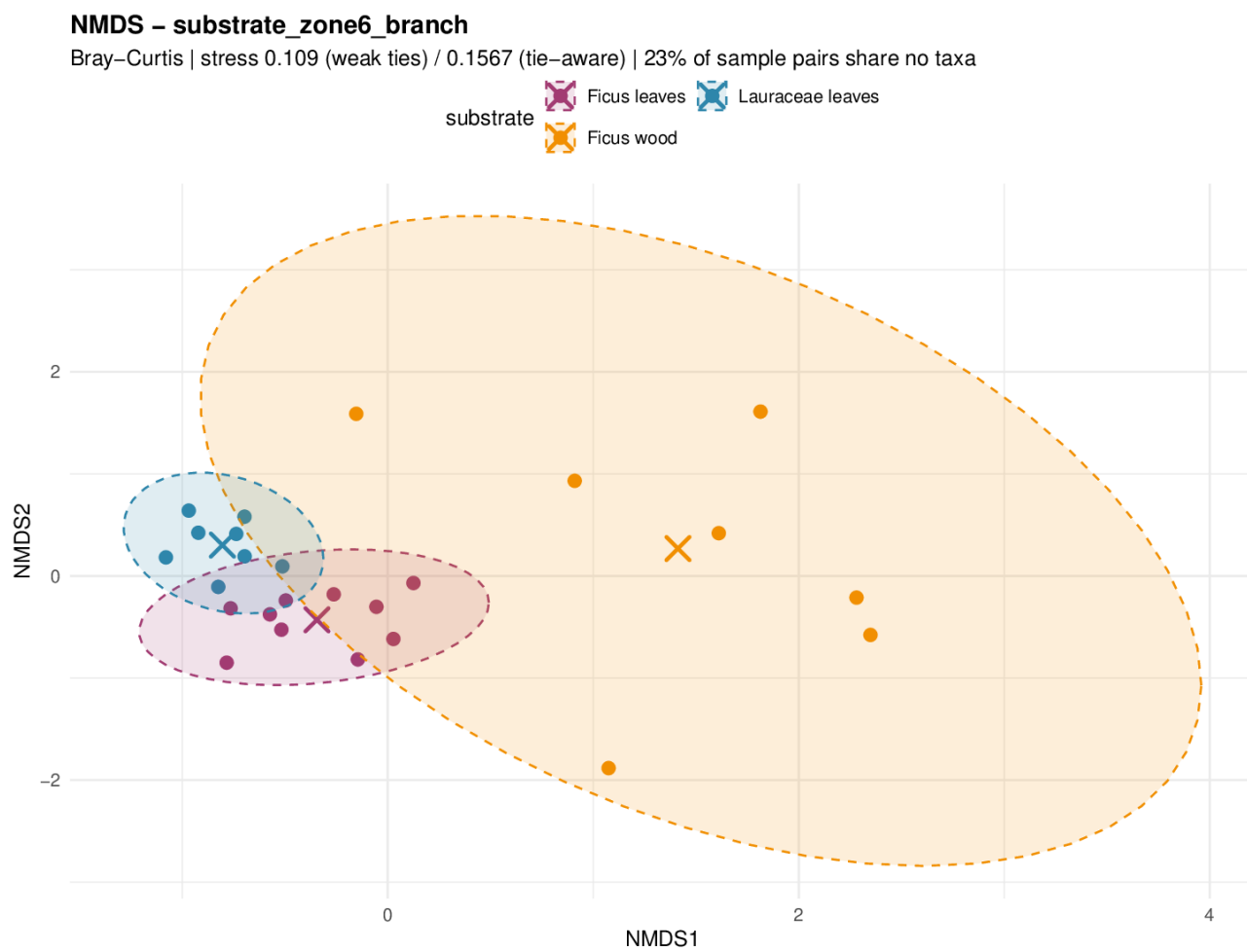


Figure 17: NMDS — substrates at zone 6

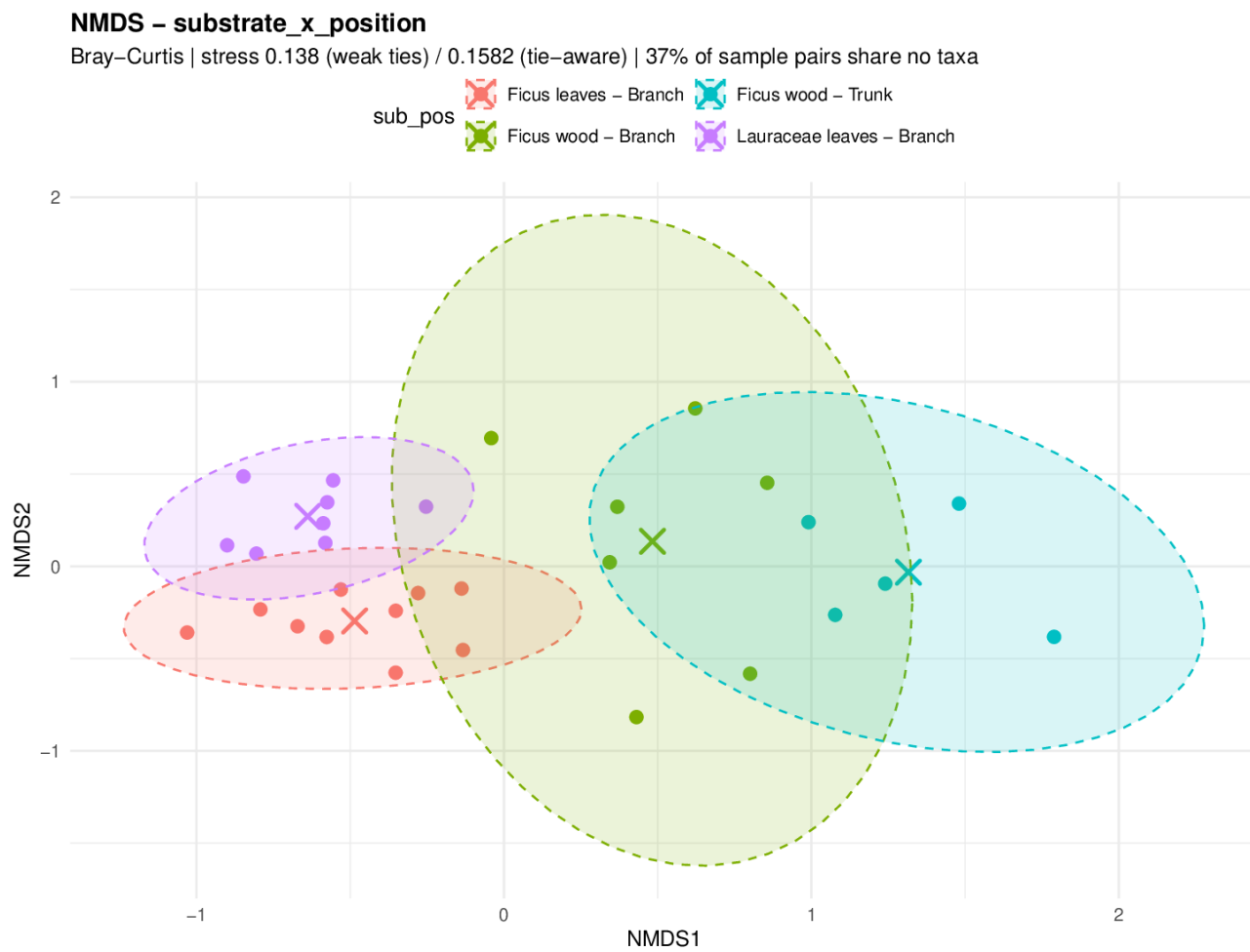


Figure 18: NMDS — substrate × position

confirmation in Section D3).

F. Branch & Trunk Sampling Orientation

Why. Different faces of a tree experience very different microclimates — sun exposure, temperature, how long the surface stays wet after rain, prevailing wind. If that matters to endophytes, communities should differ systematically between compass bearings. This is the first LOT2 analysis able to test it, because the orientation column only became usable with the re-issued samples workbook (see **Sampling orientation** above).

F1. What was actually sampled

Orientation is an **observational**, not a designed, factor here: bearings were recorded for whichever branches were reachable, so replication is uneven and, for some substrate/bearing combinations, very thin. That constrains how much the tests below can detect, and it is the single most important caveat for this section.

Substrate	Position	Orientation	Sampling units	Isolates	ITS taxa
Ficus leaves	Branch	E	2	51	34
Ficus leaves	Branch	N	2	49	30
Ficus leaves	Branch	S	2	53	43
Ficus leaves	Branch	W	3	62	46
Ficus wood	Branch	E	2	32	21
Ficus wood	Branch	N	1	11	9
Ficus wood	Branch	S	2	10	9
Ficus wood	Branch	W	2	32	24
Lauraceae leaves	Branch	E	1	2	2
Lauraceae leaves	Branch	N	4	109	63
Lauraceae leaves	Branch	NW	4	108	66

Unresolved (excluded from this section): 49 Ficus leaves branch, 82 Ficus wood trunk.

Three consequences follow, and they shape every result below:

- Only **Ficus leaves** and **Ficus wood** were sampled on all four cardinal bearings. **Lauraceae leaves** were only ever collected from northern and north-western faces (plus two isolates from one eastern unit), so the Lauraceae cannot contribute to a full-compass comparison.
- **Ficus trunk wood carries no orientation at all**, so this section is effectively a *branch*-level analysis.
- With 25 orientation-level units spread over 5 bearings, most groups hold **1-4 replicates**. Tests on a single substrate (7-9 units) have little power: a null result here means *no effect was detectable*, not *no effect exists*.

F2. Does orientation structure the community?

Same test battery as the substrate analyses — PERMANOVA, ANOSIM, PERMDISP — run on the orientation-level sampling units at ITS-genotype resolution.

Analysis	Units	Groups	F	R ²	p	ANOSIM R	ANOSIM p	Disp. p	Indic.
Orientation - all substrates	25	5	1.2780	0.2036	0.073	0.0472	0.218	0.105	0

Analysis	Units	Groups	F	R ²	p	ANOSIM R	ANOSIM p	Disp. p	Indic.
Orientation - Ficus leaves	9	4	1.2002	0.4186	0.154	0.0000	0.485	—	0
Orientation - Ficus wood	7	4	0.8327	0.4544	0.839	-0.2593	0.854	—	0
Orientation - Lauraceae leaves	9	3	2.6648	0.4704	0.048	0.3993	0.039	—	0
Substrate x orientation	25	11	2.1595	0.6067	0.001	0.6098	0.001	—	0
Sun aspect - all substrates	25	3	1.8643	0.1449	0.007	0.1705	0.016	0.068	0
Sun aspect - Ficus leaves	9	3	1.2741	0.2981	0.122	-0.1181	0.639	—	—
Sun aspect - Ficus wood	7	3	0.9982	0.3329	0.462	0.1122	0.365	—	—
Sun aspect - Lauraceae leaves	9	2	4.1375	0.3715	0.126	1.0000	0.126	—	—

F / *R*² / *p* are PERMANOVA; “Disp. *p*” is PERMDISP; “Indic.” counts significant indicator taxa.

PERMDISP shown as — where the smallest group holds fewer than 3 sampling units. A centroid computed from one or two points has a degenerate spread, which inflates the dispersion F ratio to meaningless magnitudes; those tests are written to the *_betadis-per.txt files with a warning but are not reported as numbers.

Orientation on its own: F = 1.278, R² = 0.2036, p = 0.073 — not statistically significant. Within each substrate taken separately: Ficus leaves p = 0.154; Ficus wood p = 0.839; Lauraceae leaves p = 0.048. One result is nominally significant: **Lauraceae leaves (p = 0.048, Holm-adjusted p = 0.144).** It does not survive correction for the 3 per-substrate tests. It also rests on thin, lopsided sampling. The Lauraceae units are 1 E, 4 N, 4 NW, so the test is essentially north vs north-west: two bearings 45° apart, both sun-facing. Treat it as a lead for targeted sampling, not a finding. For both *Ficus* substrates the ANOSIM R is zero or negative: units from *different* bearings are no less similar than units from the *same* bearing, which is the signature of no orientation structure at all.

Interpretation. The bearings overlap almost completely, with centroids piled on top of one another. These are ITS-level ordinations, so — as everywhere in this report — the maps are degenerate and carry no weight on their own; the conclusion rests on the PERMANOVA and ANOSIM results above, and on the higher-rank ordinations in **F7**, which agree. Whatever separates these communities, it is not which side of the tree the material came from.

F3. Separating orientation from substrate

Two of the pooled analyses above **do** come out significant — *substrate x orientation* and *sun aspect* — and both need care, because orientation is partly confounded with substrate in this dataset (the Lauraceae units are almost all north/north-west facing). A marginal (type-III) PERMANOVA asks what each factor explains **once the other is accounted for**:

PERMANOVA - variance partitioning, Bray-Curtis distance
Only sampling units with a reconciled orientation are used.
Permutations: 999

== Marginal (type III): each term adjusted for the other ==
Permutation test for adonis under reduced model
Marginal effects of terms
Permutation: free
Number of permutations: 999

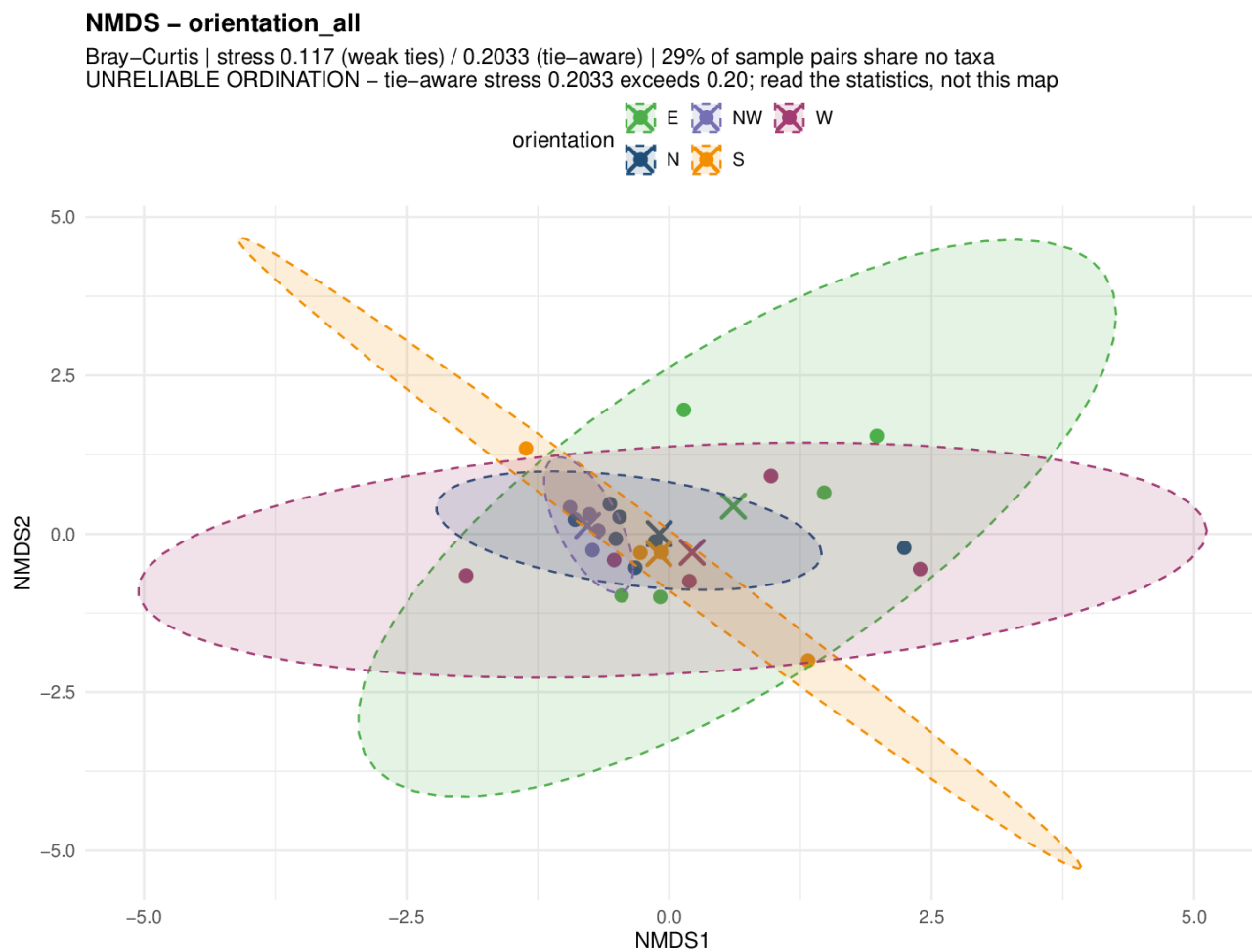


Figure 19: NMDS — orientation, all substrates

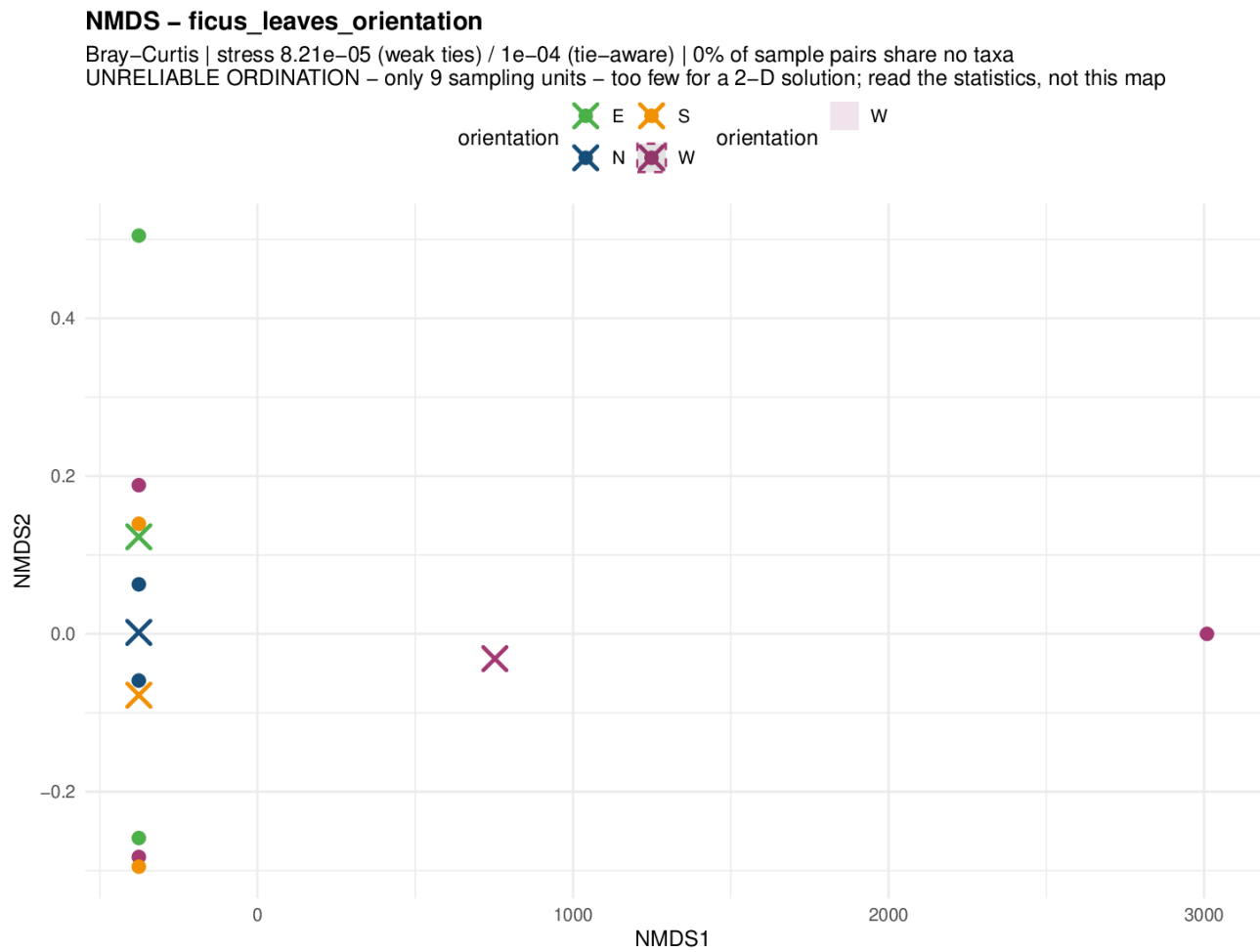


Figure 20: NMDS — orientation within Ficus leaves

```

adonis2(formula = comm_orient ~ substrate + orientation, data = meta_orient, permutations = 999, metho
      Df SumOfSqs      R2      F Pr(>F)
substrate  2   1.9763 0.22489 3.5413 0.001 ***
orientation 4   1.2370 0.14076 1.1083 0.289
Residual   18   5.0226 0.57154
Total      24   8.7878 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

== Sequential (type I): substrate entered first ==
Permutation test for adonis under reduced model
Terms added sequentially (first to last)
Permutation: free
Number of permutations: 999

```

```

adonis2(formula = comm_orient ~ substrate + orientation, data = meta_orient, permutations = 999, metho
      Df SumOfSqs      R2      F Pr(>F)
substrate  2   2.5282 0.28770 4.5304 0.001 ***
orientation 4   1.2370 0.14076 1.1083 0.289
Residual   18   5.0226 0.57154
Total      24   8.7878 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

== Marginal (type III): substrate + sun aspect ==
Permutation test for adonis under reduced model
Marginal effects of terms
Permutation: free
Number of permutations: 999

```

```

adonis2(formula = comm_orient ~ substrate + sun_aspect, data = meta_orient, permutations = 999, method
      Df SumOfSqs      R2      F Pr(>F)
substrate  2   2.0078 0.22847 3.6462 0.001 ***
sun_aspect 2   0.7531 0.08570 1.3676 0.081 .
Residual   20   5.5065 0.62661
Total      24   8.7878 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

== Sampling units per substrate x sun aspect ==
(empty cells show where aspect is confounded with substrate)

```

	Lateral (E/W)	Shaded (S)	Sun-facing (N)
Ficus leaves	5	2	2
Ficus wood	4	2	1
Lauraceae leaves	1	0	8

Substrate: $R^2 = 0.22489$, $p = 0.001$. Orientation: $R^2 = 0.14076$, $p = 0.289$. Substrate holds up; orientation does not. The apparently strong *substrate* $\times$ *orientation* result ($R^2 \approx 0.6067$) is therefore the substrate effect re-expressed through a factor that happens to encode it, not evidence that bearing matters.

Sun aspect LOT2 was collected in **Peru, i.e. the southern hemisphere**, where the *northern* face of a tree is the sun-exposed one and the southern face the shaded one. Grouping bearings into **sun-facing (N/NE/NW)**, **shaded (S/SE/SW)** and **lateral (E/W)** gives a more direct microclimate proxy than the raw compass, and coarser groups mean better replication.

Pooled across substrates this is significant ($R^2 = 0.1449$, $p = 0.007$, PERMDISP $p = 0.068$, so not a dispersion artefact) — but the confounding is severe: of the 25 orientation units, **all but one Lauraceae unit is sun-facing**, while Ficus leaves and wood are mostly lateral. Adjusting for substrate, **sun aspect gives $R^2 = 0.08570$, $p = 0.081$ — not statistically significant**. Within each substrate individually it is likewise non-significant. The pooled result is substrate wearing an aspect label.

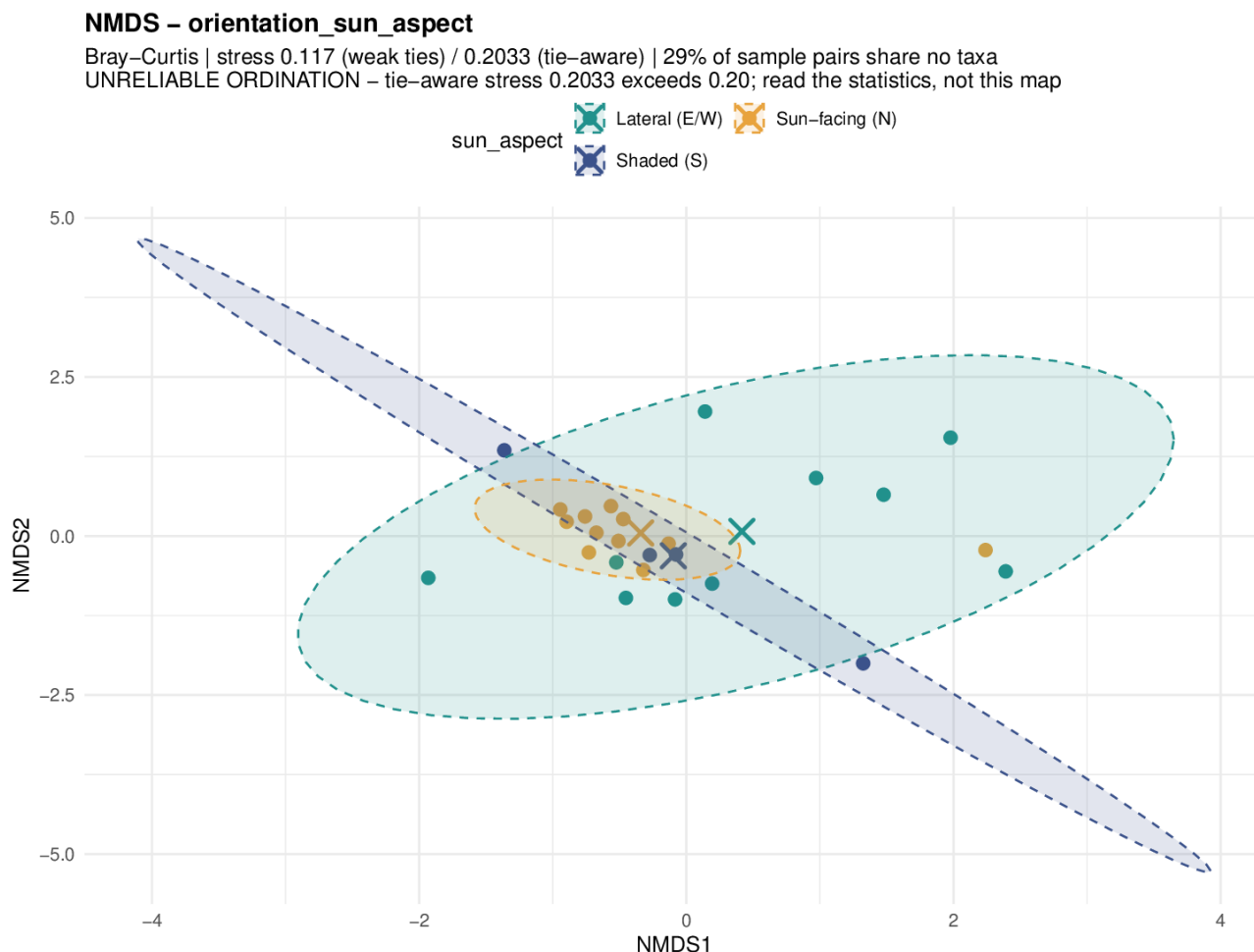


Figure 21: NMDS — sun aspect

F4. A directional gradient rather than discrete bearings

Treating the compass as four or five **discrete classes** spends a lot of degrees of freedom on a dataset this small. An alternative is to treat the bearing as the **circular variable** it is: decomposing it into $\cos(\text{bearing})$ (the north-south axis) and $\sin(\text{bearing})$ (the east-west axis) tests for a community that changes *smoothly* around the tree using only 2 df, which is more powerful at this level of replication.

Analysis	Units	Bearings	N-S axis R^2	p	E-W axis R^2	p	Joint R^2	Joint p
All substrates	25	5	0.0628	0.065	0.0375	0.534	0.1051	0.120
Ficus leaves	9	4	0.1304	0.442	0.1234	0.522	0.2538	0.423
Ficus wood	7	4	0.1937	0.343	0.1214	0.884	0.3151	0.624
Lauraceae leaves	9	3	0.1993	0.059	0.1588	0.057	0.4704	0.048

Interpretation. The only nominally significant result is the joint gradient in Lauraceae leaves ($p = 0.048$), out of 12 uncorrected tests in this table. The Lauraceae need a specific warning. They span only N and NW plus a single eastern unit, so any “gradient” is fitted over a $\sim 45^\circ$ arc plus one point. With exactly three bearings, the 2-df joint test is mathematically the same test as the three-group PERMANOVA in F2 ($R^2 = 0.4704$ in both), so it is not independent confirmation.

F5. Diversity per orientation

Community *structure* may not differ while *diversity* still does — a more exposed face could simply support fewer taxa. Richness depends strongly on how many isolates each group contributed, so richness is also reported **rarefied to a common isolate count**.

Orientation	S	N	H'	1-D	J'	Rarefied S (to 63)
E	57	85	3.8174	0.9694	0.9442	45.10
N	94	169	4.2844	0.9817	0.9430	46.54
NW	66	108	3.9516	0.9738	0.9432	44.48
S	52	63	3.8845	0.9776	0.9831	52.00
W	66	94	4.0425	0.9783	0.9649	48.68

Sun aspect	S	N	H'	1-D	J'	Rarefied S (to 63)
Lateral (E/W)	114	179	4.4573	0.9816	0.9411	49.58
Shaded (S)	52	63	3.8845	0.9776	0.9831	52.00
Sun-facing (N)	132	277	4.5091	0.9831	0.9235	46.76

Interpretation. Raw richness tracks sampling effort almost exactly — the most-sampled bearing is also the richest — but once rarefied to a common 63 isolates the bearings are within a few taxa of each other (44.48-52 taxa). Evenness (Pielou J') is uniformly high, as everywhere else in this dataset. There is no diversity gradient around the tree to match the absent compositional one.

The per-substrate rarefaction target is pulled down to a handful of isolates by the smallest Ficus wood groups, so `tables/orientation_alpha_diversity_by_substrate.csv` should be read as indicative only. The pooled table above is the more reliable of the two.

F6. Taxon sharing between orientations

Occurs in ... orientations	ITS genotypes
1	182
2	35
3	17
4	8

Interpretation. Most genotypes (182 of 242) were found on a single bearing. On its own that looks like strong orientation fidelity, but it is the expected consequence of **rarity plus thin sampling**: this community is dominated by singletons and doubletons (see the rank-abundance curves in Section B1), and a taxon seen once can only ever be recorded at one bearing. The multivariate tests above, which use the whole community at once rather than taxon-by-taxon presence, find no such structure — and they are the trustworthy reading.

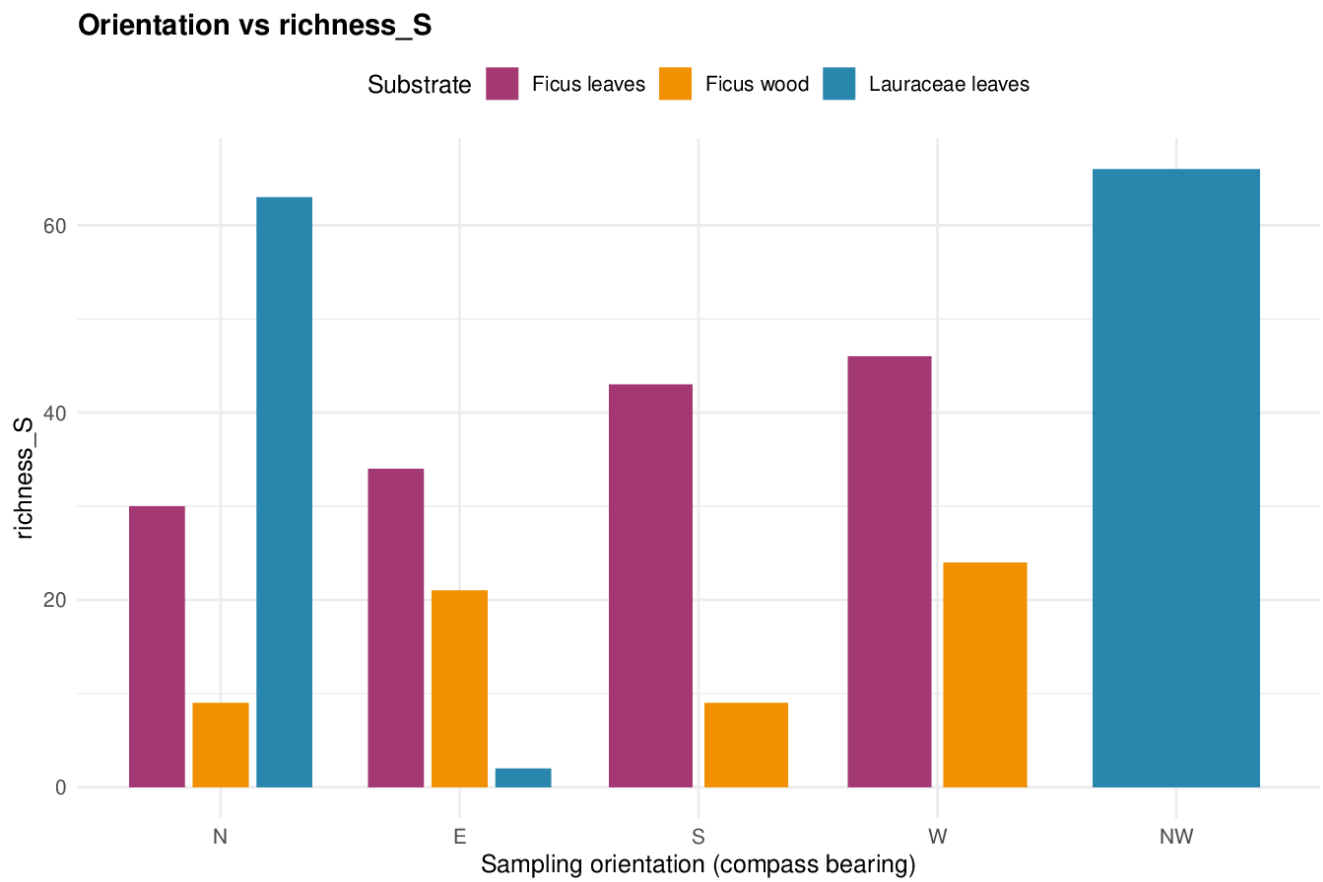


Figure 22: Richness by orientation and substrate

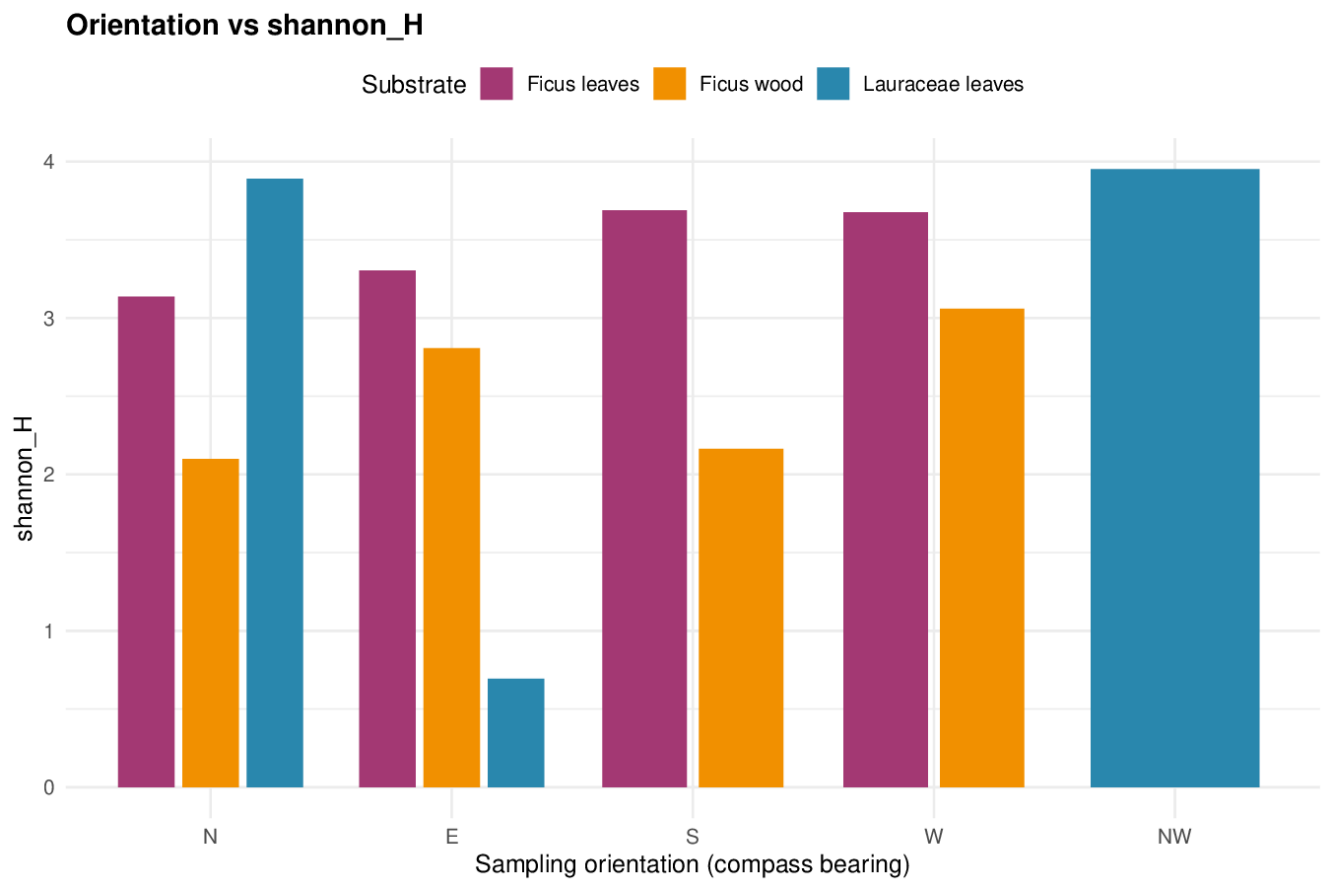


Figure 23: Shannon diversity by orientation and substrate

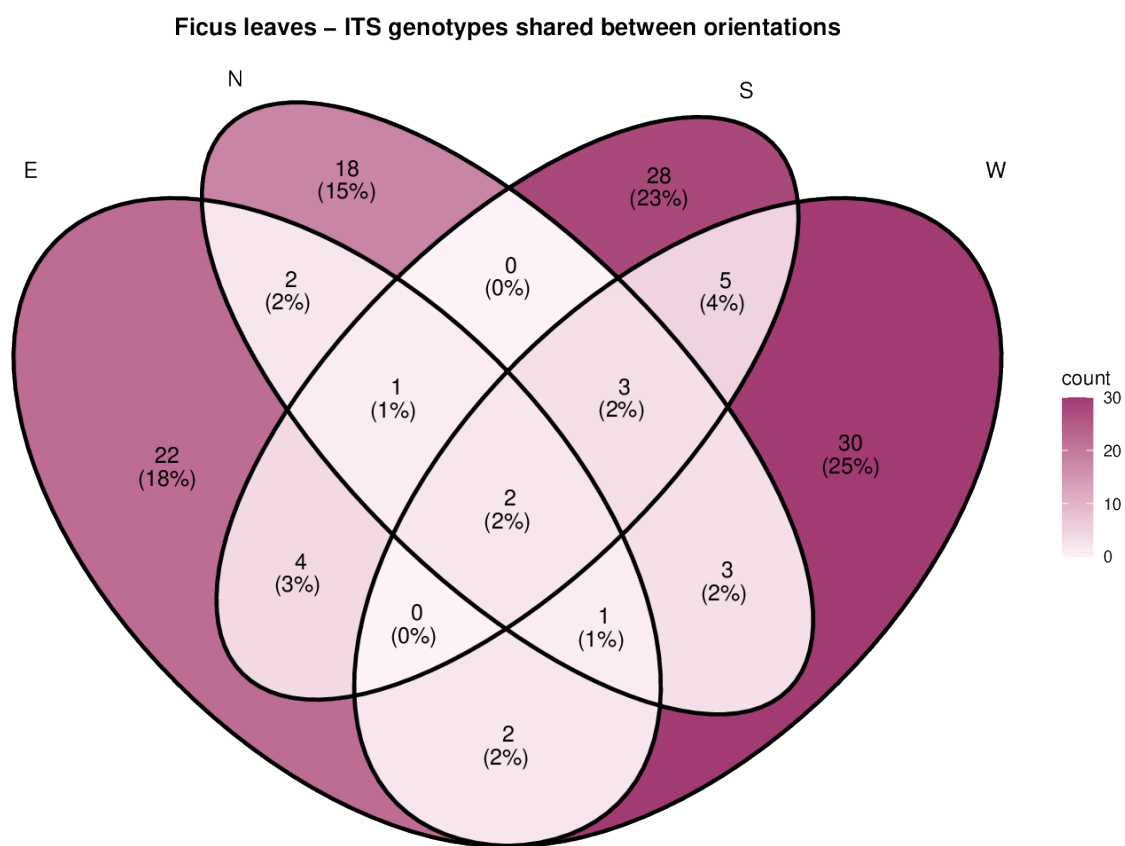


Figure 24: Shared genotypes between orientations — Ficus leaves

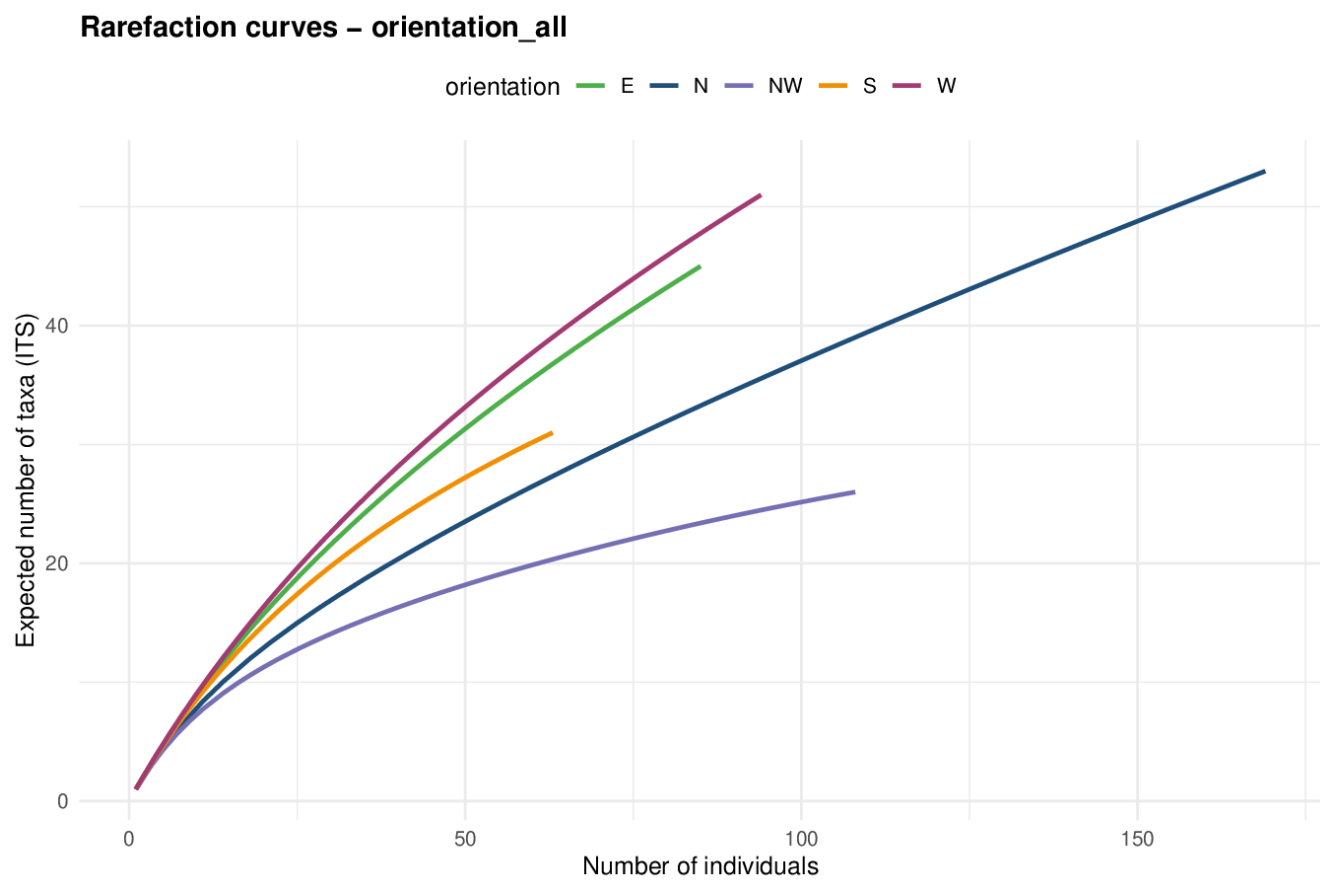


Figure 25: Rarefaction by orientation

F7. Orientation across taxonomic ranks

As in Section D, the comparison is repeated at every rank, in case a bearing effect exists among broader taxonomic groups but is hidden by genotype-level noise.

Rank	Taxa	No-taxa pairs	Stress	Tie-aware	F	R ²	p	ANOSIM R	ANOSIM p	PERMDISP p
phylum	4	0%	0.0055	0.0152	1.2624	0.2016	0.280	0.0707	0.140	0.229
class	8	0%	0.1018	0.0746	1.8154	0.2664	0.079	0.1598	0.026	0.093
order	22	4%	0.1294	0.1352	1.1430	0.1861	0.322	0.0409	0.240	0.035
family	28	12%	0.1379	0.1328	1.2490	0.1999	0.199	0.0474	0.203	0.047
genus	34	20%	0.1747	0.1382	1.4412	0.2237	0.096	0.0778	0.123	0.022
species	60	32%	0.1023	0.1978	1.1114	0.1819	0.286	0.0387	0.242	0.237
otu_97	121	29%	0.1165	0.2033	1.2780	0.2036	0.073	0.0472	0.218	0.105
its_taxon	242	52%	0.0001	0.2360	1.1661	0.1891	0.077	0.0718	0.138	0.099

$F / R^2 / p$ are PERMANOVA; “No-taxa pairs” is the share of sample pairs with no taxon in common; “Tie-aware” is the stress refitted with strong ties — the honest one to judge by.

Interpretation. Orientation is non-significant at **every** taxonomic rank, so the null result is not an artefact of working at ITS resolution.

Full results: tables/orientation_*, tables/*_orientation_* and tables/substrate_x_orientation_*.
Summary: tables/orientation_summary.csv.

F8. What this section concludes

Sampling orientation does not detectably structure the LOT2 endophyte communities. It is non-significant pooled, at every taxonomic rank, as a coarse sun-exposure grouping, and as a smooth directional gradient pooled across substrates. The one exception is Lauraceae leaves ($p = 0.048$, Holm-adjusted $p = 0.144$), which does not survive correction across substrates and compares branches facing N and NW (1 E, 4 N, 4 NW units). The two significant pooled results both dissolve once substrate is accounted for (orientation adjusted for substrate: $R^2 = 0.14076$, $p = 0.289$; sun aspect adjusted for substrate: $R^2 = 0.08570$, $p = 0.081$).

How firm is that? Firm enough to report, but it is a negative result from an unbalanced observational factor with 1-4 replicates per group. It rules out an orientation effect of the size seen for substrate ($R^2 \approx 0.33$); it does not rule out a small one. Making that test properly would need balanced sampling of all four bearings within each substrate — worth specifying in advance if a future campaign wants to answer this question rather than check it.

Abundance & Composition Plots (Incertae sedis retained)

Like the relative-abundance bars in B2, the plots below **keep the *Incertae sedis* isolates** as an explicit pooled category, so they present the complete isolate census without hiding unidentified material. The category is grey and placed last on every chart, with its share of each substrate printed on the chart.

plots/abundance_pooled_incertae_sedis/

Horizontal grouped bar charts of **absolute isolate counts** per taxon, split by substrate. Unresolved taxa are collected into an *Incertae sedis* row, pinned to the top on a grey band, with each bar labelled by its count and share of that substrate’s isolates.

Fungal isolate abundance by genus

Incertae sedis (grey band, top row) = unknown or unresolved at this rank.

Share of isolates: Lauraceae leaves 11.4% (25), Ficus leaves 20.5% (54), Ficus wood 20.4% (34)

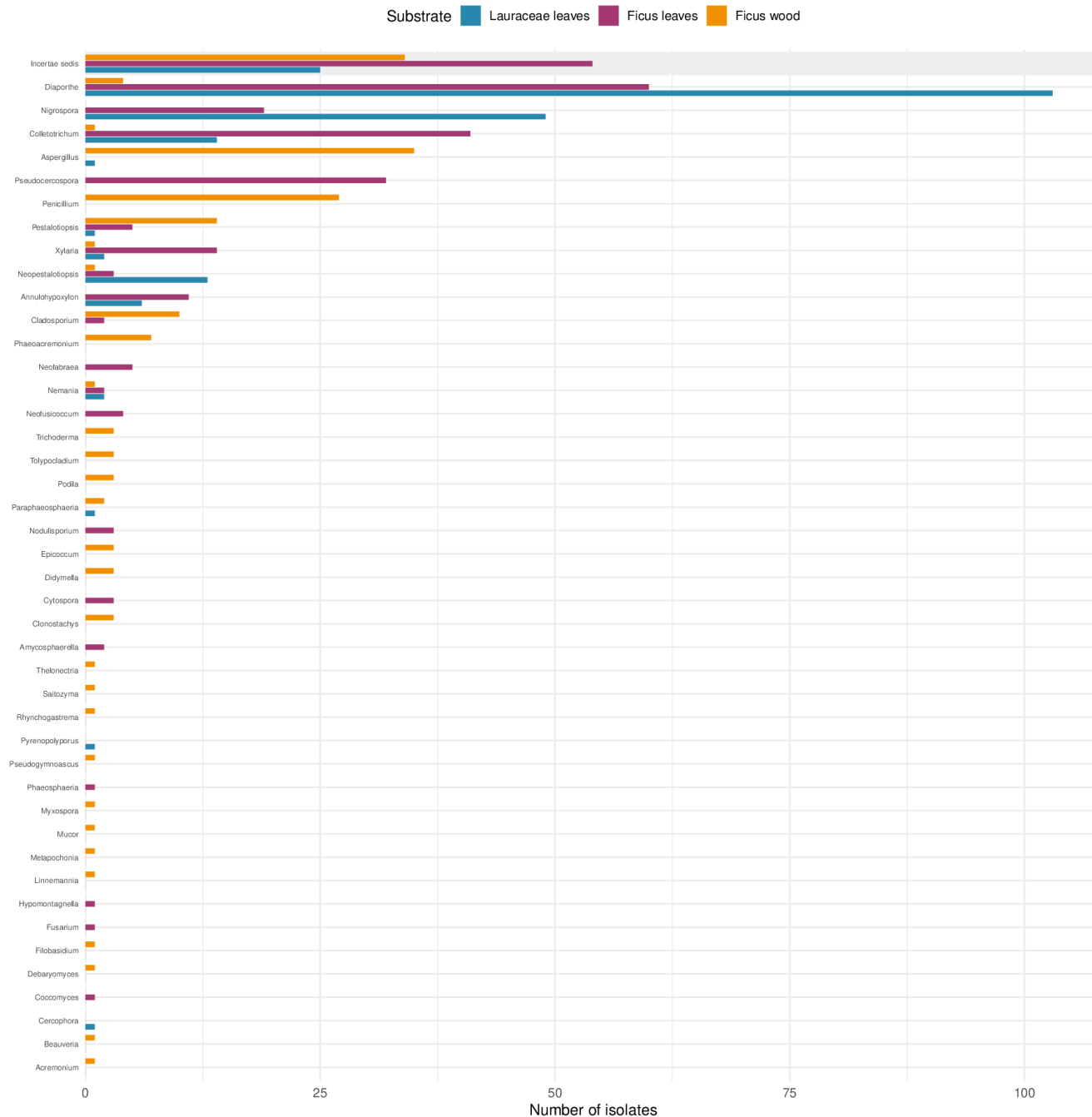


Figure 26: Abundance by genus (pooled)

plots/pie_charts/

Proportional composition of each substrate; the grey *Incertae sedis* slice shows how much of each community remains unidentified at that rank, and its share is given under each pie's title.

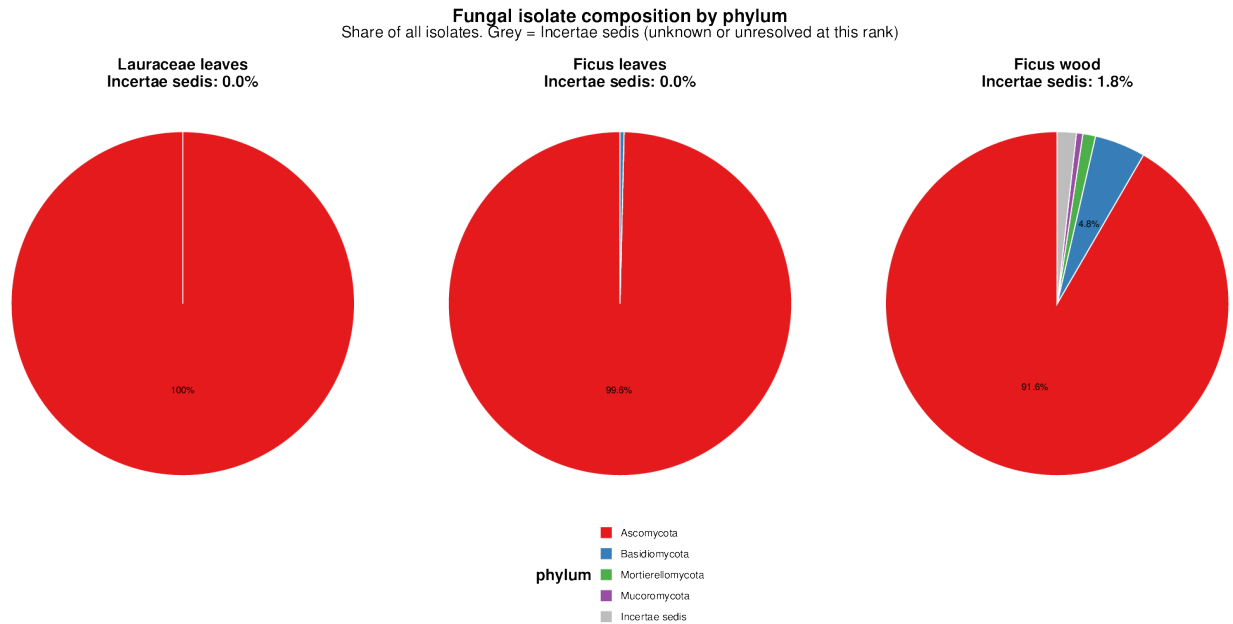


Figure 27: Pie chart by phylum

Conclusion

1. **Substrate is the primary driver of community structure.** The three substrates host significantly different fungal communities (PERMANOVA $p = 0.001$, $R^2 \approx 0.33$), and this holds at **every taxonomic rank** — the effect is in fact strongest around **class level** ($R^2 \approx 0.57$). The two leaf substrates are more similar to each other than to wood at every rank (Section D1), i.e. **tissue type (leaf vs wood)** is the strongest split, with **host identity (Ficus vs Lauraceae leaves)** adding a secondary but significant effect ($p = 0.001$).
2. **Tree height has at most a weak effect.** Treated as six discrete zones, height structures Ficus-wood communities only at **class level** ($p = 0.017$) and at no other rank (Section D2); with a single sampling unit per trunk zone, that test has little power. When the wood is instead split simply into **trunk vs branch**, a modest but **statistically significant** difference emerges ($p = 0.003$, $R^2 \approx 0.19$): branch wood carries a somewhat distinct community from trunk wood, but this coarse contrast explains far less variation than substrate does.
3. **The substrate signal is real, not a sampling-height artefact.** Even when the comparison is restricted to zone 6 alone (where all substrates co-occur), substrates remain **statistically significant** ($p = 0.001$).
4. **Which side of the tree the material came from does not matter.** Now that the field branch codes have been reconciled with the project database, sampling orientation could be tested for the first time (Section F, 519 of 650 isolates). It is non-significant pooled ($p = 0.073$), at every taxonomic rank, as a sun-exposure grouping and as a smooth directional gradient. The one exception is Lauraceae leaves ($p = 0.048$, Holm-adjusted $p = 0.144$), which does not survive correction across substrates and compares branches facing N and NW (1 E, 4 N, 4 NW units). The two pooled contrasts that do come out significant — *substrate x orientation* and *sun aspect* — both vanish once substrate is accounted for (orientation adjusted for substrate: $R^2 = 0.14076$,

